

Equity in Transition: Analyzing the Distributional Impacts of Low-Carbon Energy Technology Subsidies

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Abstract

I study the distributional and welfare effects of low-carbon energy technology subsidies in the context of US residential rooftop solar adoption. Although these subsidies are often viewed as regressive because wealthy households capture most of the benefits, I argue that this assessment changes once learning-by-doing spillovers are incorporated. Using installation-level data, I provide new empirical evidence on cost declines from cumulative adoption and estimate learning elasticities to discipline the model. I then develop a heterogeneous-agent DSGE model with incomplete markets, irreversible technology adoption, and adoption-driven cost reductions. Calibrated to US data, the model quantifies how subsidies shape adoption patterns, transitional dynamics, and heterogeneous welfare across the income-wealth distribution. Preliminary results suggest that, while pecuniary transfers appear regressive, the technological externalities improve welfare for all, leaving current policy less regressive. The framework is used to evaluate alternative subsidy designs and financing strategies—such as income-capped credits, emissions taxes, and debt financing—highlighting trade-offs between efficiency, equity, and political feasibility.

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1 Introduction

One concern about low-carbon (“clean”) technology subsidies is that high-income households claim them more due to high initial costs. For example, on average, the top income quintile in the United States (US) received more than 50% of investment tax credits for residential energy efficiency improvements since introducing these credits, according to the Internal Revenue Service’s Statistics of Income data. If these subsidies that mainly go to the richer households are funded by the general taxpayer revenues, then they are pecuniary transfers from the poor to the rich households. Such transfers are perceived as regressive in monetary terms, and as such, appear inequitable and raise discontent among the populace.

At the same time, if we consider both the environmental and learning-by-doing external economy effects alongside the general equilibrium effects, these policies that subsidize the richer households may have greater welfare benefits for poorer households than for richer ones. For example, the health co-benefits of reducing local air pollutants associated with decreasing greenhouse gas emissions are larger for the poorer households who are more commonly exposed to air pollution. Since wealthy households account for the majority of residential emissions, speeding up their transition to low-carbon technologies will yield the most environmental benefits. Moreover, subsidizing early adopters who are likely to be wealthy, could reduce the high adoption costs at an accelerated rate than otherwise, making adoption feasible for later adopters who are likely to be less wealthy.

Motivated by this regressivity concern, I ask: How is low-carbon technology adoption related to household income in the US? Focusing on residential rooftop solar panel adoption in the US, I show a positive correlation between income and aforementioned adoption rates. If the richer households receive more of these subsidies then what are the distributional welfare implications of existing residential clean technology subsidies in the US over the joint income and wealth distribution? By calibrating a model to reflect the US economy and policy in the early days of the energy transition, I can quantify the equity and efficiency of the recent energy transition policies that promote adoption of residential rooftop solar panels.

Finally, I ask what is the best policy mix to achieve different policy objectives, such as emissions reduction, welfare maximization, and transition speed, together with majority support. I use my calibrated model to evaluate the optimal policy mix under these different objectives and their majority support. First, I evaluate the overall welfare effects and political feasibility of different financing strategies for a uniform subsidy payment schedule through policies including the following: a uniform labor income tax, a progressive labor income tax, a corporate income tax, an emissions tax, and finally, government debt. In a second set of

experiments, I evaluate the effectiveness of different subsidy programs’ ability to meet the outlined policy objectives with an example being an income-capped subsidy program. These comparative policy experiments can inform policy design and implementation of residential energy efficiency improvement tax credits in the US and other countries.

Versions of the questions posed here have been addressed in literature. Most of the previous research falls into two categories: (i) empirical analyses of the distributional effects of environmental policies, and (ii) the macroeconomic studies on the unequal economic consequences of climate change policies.

Borenstein and Davis (2024) have well documented the adoption patterns for various low-carbon energy technologies and tax credit receipts for these technologies across income groups in the US. However, their study neither goes beyond descriptive facts nor quantifies dynamic general equilibrium effects, such as the indirect effects on relative prices, or the nonpecuniary effects such as the health co-benefits of reducing local air pollutants. Vona (2023) summarizes the multidimensional impacts that climate policies have on well-being which have been documented by various researchers and provides an overview of the empirical literature. The consensus of this empirical literature is that subsidies to clean and energy-efficient equipment exacerbate the regressivity of climate policies in the presence of financial constraints by benefiting the wealthier households at the expense of less wealthy ones.

I will contribute to these empirical analyses of who benefits from clean technology subsidies by quantifying the overall distributional effects of these policies while taking into account multidimensional effects using a model calibrated to reflect the US economy. Specifically, I incorporate two additional indirect distributional effects of clean technology subsidies into a general equilibrium model: (i) the health co-benefits of reducing local air pollutants associated with reducing greenhouse gas (GHG) emissions, and (ii) the learning-by-doing spillover effects of early adopters. Reducing GHG emissions has well-known positive effects in terms of health co-benefits such as reducing local air pollutants and improving health outcomes. Banzhaf, Ma, and Timmins (2019) summarizes the large body of economic literature that documents a strong positive correlation between ambient air pollution, poverty, and race, i.e. the so-called environmental justice gap. The second dimension that I aim to incorporate in my analysis is the learning-by-doing spillover effects from early adopters, who are likely to be wealthier. Gao, Rai, and Nemet (2022) is one of few empirical economic analyses which finds that learning through economies of scale in production leads to cost reductions, not only in hardware costs, but also in non-hardware costs in US residential solar installations.

There are several quantitative macroeconomic studies that analyze the distributional effects of climate policies in a general equilibrium framework. The majority of these studies focus on the distributional effects of carbon pricing instead of low-carbon technology subsidies

like my paper does. Känzig (2023), Benmir and Roman (2022), Fried, Novan, and Peterman (2024), Fried, Novan, and Peterman (2018), and Belfiori, Carroll, and Hur (2024) are some recent studies that analyze the distributional effects of carbon pricing.

Känzig (2023) finds that a carbon tax is regressive using institutional features of the European emissions trading system and high-frequency data. Benmir and Roman (2022) studies the economic consequences of carbon pricing that achieves the net-zero emissions target in the US by 2050 and find that the policy induces large redistribution of income and wealth from poor to rich households. Fried, Novan, and Peterman (2024) study the welfare and inequality implications of different ways to return carbon tax revenue back to households and find that the welfare-maximizing rebate uses two-thirds of carbon tax revenue to reduce the distortianary capital income tax while using the remaining one-third to increase the progressivity of the labor income tax. In an earlier work Fried, Novan, and Peterman (2018) study how different approaches for recycling carbon tax revenue affect the welfare of current and future generations. Their work highlights the importance of not only long-run outcomes, but also the transitional welfare effects of how carbon tax revenues are recycled. Using household expenditure and emissions data, Belfiori, Carroll, and Hur (2024) document that low-income households have higher emissions per dollar spent than high-income households, making a carbon tax regressive. Overall, the literature shows that carbon pricing is regressive, but the regressivity can be mitigated by recycling the carbon tax revenue in a progressive way.

Kuhn and Schlattmann (2024) is the closest work to this project in that they develop a quantitative life-cycle model with heterogeneous adoption rates of carbon-neutral commitment goods by income to quantify the reduction-redistribution trade-off of different policy mixes. Lanteri and Rampini (2025) is another similar macroeconomic study of adoption of clean technologies, but instead by heterogeneous firms, the study uses a dynamic general equilibrium model of firm dynamics and clean technology adoption with financial constraints. They find that financially constrained firms optimally invest in dirty new technologies as well as in older technologies, resulting in a positive relation between firm size and energy efficiency. They highlight that their proposed model could be a laboratory for studying the distributional effects of environmental policy across firms, however they leave these exercises for future work. The computational methods I propose in this paper could easily be extended to study heterogeneous firms rather than households. This extension has a high priority on my research agenda.

Through my review, I have yet to find a quantitative macroeconomic study that includes the health co-benefits and learning-by-doing spillover effects in the distributional analyses of climate policies for heterogeneous households.

Although job reallocation is a significant distributional consequence of transition to low-carbon technologies, this paper will not consider the labor market effects of the transition. Graham and Knittel (2024) is the most recent empirical study that map the spatial distribution of employment vulnerability to the energy transition in the US. Cavalcanti, Hasna, and Santos (2021) is a quantitative macroeconomic analysis of the impact of climate change mitigation policies on labor reallocation across sectors. My proposed quantitative model for distributional analysis using a heterogeneous agent model could be extended to include the labor reallocation effects in a future paper.

I will answer the three questions that I posed above using a combination of empirical and quantitative methods. Each answer will form the basis of the answer to subsequent questions. The remainder of the paper is structured as follows. In Section 2, I summarize the data that motivate the research questions and provide background for the model. In Section 3, I outline the structural model that I use to answer my research questions. In Section 4, I describe the complete characterization of the model used for quantitative analysis, its calibration and fit to the data, and present the baseline model simulations. In Section 5, I conduct policy experiments to evaluate the effectiveness of different policy mixes in achieving the outlined policy objectives and majority support. Finally in Section 6, I conclude and discuss the implications of the results for policy design and implementation.

2 Data and Empirical Motivation

To motivate my research questions, it is essential to understand the distributions of residential energy consumption and emissions, adopters of on-site solar power generation, the receipts of residential energy credits, ambient air pollution exposure, and energy expenditure shares across income quintiles in the US. I summarize these distributions in Table 1 using various cross-sectional data sources from the US in 2015. I focus on 2015 because it is the earliest year for which data on all these variables is available.

First, using data from the US Census Bureau’s (2023) 2015 American Community Survey (ACS) 5-Year Estimates, I construct income quintiles based on the upper income limits of quintiles summarized in Table B19080. For 2015, these limits are \$17,929, \$35,583, \$62,600, \$108,429, and the lower limit for top 5 percent is \$146,778. I report the share of aggregate income for each income quintile from the US Census Bureau’s (2023) 2015 ACS Table B19082 in the first row of Table 1. The top income quintile accounts for more than half of the aggregate income in the US.

Second, I use the RECS’s (2023) 2015 Residential Energy Consumption Survey (RECS) data to calculate the share of aggregate residential energy consumption for each income

Table 1: Descriptive statistics for the income quintiles in the US

income percentile	bottom 20%	20%-40%	40%-60%	60%-80%	top 20%	top 5%
share of aggregate income	3.17	8.42	14.37	22.83	51.21	22.81
share of residential energy consumption	12.59	18.16	13.07	19.75	24.42	12.01
share of emissions from residential energy	13.43	18.36	13.89	20.20	23.01	11.05
share of rooftop solar adoptors	0.53	3.42	9.26	16.87	43.75	26.19
share of residential clean energy credits	0.48	4.11	4.08	21.75	48.99	20.59
mortality damages per capita (2020 dollars pp)	4,811	3,910	3,103	2,769	2,354	na

Notes: Reported shares and rates are in percentages, except for the mortality damage per capita values, which is in 2020 US dollars per person (pp).

quintile. Annual household income is reported as a categorical variable in the RECS data, and I group households according to the income quintile’s upper limits as closely as possible. Thus, the upper income limits for the quintiles I report from the RECS data are \$20,000, \$40,000, \$60,000, \$100,000, and the lower limit for top 5 percent is \$140,000. I use the household weights provided in the RECS data for all of my calculations. I calculate the share of aggregate residential energy consumption for each income quintile by aggregating total energy consumption in British Thermal Units (BTUs) for each group and the total, and calculating the share of each group’s total to the national total. The results reported in the second row show that the top income quintile accounts for almost a quarter of the aggregate residential energy consumption in the US.

Third, I calculate the share of emissions from residential energy consumption for each income quintile using the composition of different energy sources’ PM2.5 emissions intensity from the US Environmental Protection Agency’s (2018) Emissions & Generation Resource Integrated Database (eGRID) and each household’s consumption of each energy source from the 2015 RECS. I use the following PM2.5 emission factors in (lb/MMBTU) for each energy source: 0.219401 for electricity, 0.0004 for natural gas, 0.215567 for fuel oil and kerosene, 0.0004 for propane, 0.177 for cordwood and wood pellet. I calculate the share of emissions from residential energy consumption for each income quintile by multiplying the PM2.5 emissions intensity of each energy source by the consumption of that energy source for each household, summing up the emissions for each household, and calculating the share of each group’s total to the national total. The results reported in the third row show that the share of residential emissions attributed to each group is similar to its share of residential energy consumption, but share of emissions is greater than the share of energy consumption for every group except the top quintile and the top 5 percent. In other words, the top income quintile and the top 5 percent have a lower share of emissions than their share of energy consumption. As I argued in the introduction, the top income quintile is responsible for a greater share of energy consumption and emissions than their share of the population.

Next, I summarize some facts about rooftop solar panel deployment rates and the receipts of residential energy credits that cover installment of residential solar panels, across income quintiles. I use the 2015 RECS data to calculate the rooftop solar panel deployment rates for each income quintile. I calculate the deployment rate for each income quintile by dividing the number of households that generate power on-site using solar in that quintile by the total number of households that generate power on-site using solar across all income groups. The results reported in the fourth row show that share of rooftop solar adopters increases with income, with the top income quintile having having almost half of the adopters.

The US has several government-sponsored incentive programs that reduces the cost for people and businesses to use alternative energy sources. Eligible taxpayers meeting the criteria get the credit amount deducted from their total tax bill. There are two types of energy investment tax credits (ITCs) available to homeowners: the Energy Efficient Home Improvement Credit and the Residential Clean Energy Credit. Taxpayers can only use one or the other of these two residential energy tax credits in a tax year.

The Energy Efficient Home Improvement Credit offers a 30% ITC to cover some of the cost of eligible home improvements, such as alterations to exterior doors, windows, and electric or natural gas heat pumps. The Residential Clean Energy Credit (RCEC), formerly known as the Residential Energy Credit, is a ITC that was extended and renamed under the Inflation Reduction Act. The credit covers 30% of the cost of installing solar panels, solar water heaters, geothermal heat pumps, small wind turbines, geothermal heat pumps, fuel cells and battery storage technologies of at least 3 kilowatt (kW) per hour (kWh). The credit is available for both existing homes and new construction, but the home must be the taxpayer's primary residence. The RCECs were enacted as part of the Energy Policy Act of 2005 and were extended and expanded several times since then. The final extension and expansion of the RCECs was under the Inflation Reduction Act, increasing the credit back to 30%, where it is set to remain until 2032 before dropping to 26% in 2033 and then 22% in its final year, 2034. Importantly, the RCECs are non-refundable, meaning that the credit cannot exceed the taxpayer's tax liability. The non-refundability is a key feature of the RCECs that makes them regressive, as only those with a tax liability, who tend to be wealthier, can benefit from the credit.

I use the US Internal Revenue Service's ([2023](#)) 2015 Statistics of Income (SOI) data to calculate the share of total value of the RCECs issued attributed to each income group. Unfortunately, the publicly available SOI data does only provide income categories and the ranges for these categories do not align perfectly with the income quintiles I use. Therefore, I group the income categories in the SOI data to match the income quintiles as closely as possible. Specifically, the upper income limits for the income categories I report from the

SOI data are \$20,000, \$40,000, \$50,000, \$100,000, and the lower limit for top 5 percent is \$200,000. I calculate the share of the amount of RCECs claimed by each income group by dividing the total amount of residential energy credits issued to each group by the total amount of residential energy credits issued to all income groups. The results reported in fifth row show that the share of RCECs claimed by the income group increases with income, with the top income quintile receiving almost half of all residential energy credits issued in the US.

These five statistics describe that higher income households consume more energy, emit more pollutants, adopt rooftop solar panels more, and receive more residential energy credits than lower income households. Thus, having richer households adopt clean technologies faster could yield the most emissions reductions. However, as I argued in the introduction, the health co-benefits of reducing local air pollutants is greatest for the poorer households who are more commonly exposed to air pollution. In order to provide evidence for this claim, I use Dennin et al.’s (2024) latest Air Pollution Emission Experiments and Policy Analysis model, the AP4 model, to calculate the county-level mortality damages per capita for 2017. I match these mortality damages with county-level median income data from the US Census Bureau’s (2022) 2017 ACS 5-Year Estimates. The results reported in the sixth row show that mortality damages per capita are inversely related to income, with the bottom income quintile experiencing the highest damages. Thus, reducing emissions by a unit would prevent greater damages for the poorer households. These are some of the distributional facts that motivate my research questions and I will use for the calibration of my model.

2.1 Benefits of Residential Rooftop Solar Panel System Deployment

Deploying solar panels for on-site power generation has both private and social benefits. The private benefits include reduced electricity bills, increased property values, and reduced exposure to electricity price volatility. The social benefits include reduced emissions of greenhouse gases and local air pollutants, reduced strain on the electricity grid, and increased energy security. In this section, I describe these potential benefits of adopting solar panels for on-site power generation, provide descriptive evidence on the private benefits, and describe the potential social benefits that are not internalized by households.

2.1.1 Private Benefits

First, I describe the private benefits of adopting solar panels for on-site power generation. The most immediate private benefit is a reduction in household electricity expenditures. To

Table 2: Average 2015 energy expenditure shares of income groups in the US

income percentile	bottom 20%	20%-40%	40%-60%	60%-80%	top 20%	top 5%
share of energy expenditure in total expenditure	8.5	6.4	5.0	4.2	3.3	2.2

Note: Reported shares are in percentages.

illustrate the magnitude of this accounting effect, I use the 2020 Residential Energy Consumption Survey (RECS). This dataset includes information on annual household electricity expenditures and consumption, as well as an indicator for whether the household has on-site solar generation.

Unsurprisingly, households with on-site solar report substantially lower grid electricity expenditures. For example, a simple regression of electricity expenditures on a solar indicator (presented in Appendix Table 6) shows that solar households spend roughly \$700 less annually on electricity. The average annual electricity expenditure in the RECS data is about \$1,400, so this represents a substantial reduction. However, this estimate should not be interpreted as a causal effect: it simply reflects the accounting identity that households generating their own electricity purchase less from the grid. Because solar adoption is endogenous to household characteristics and policy incentives, these regressions are not identified. I therefore report them only to illustrate magnitudes and relegate the tables to the appendix.

The key point for calibration is that these private pecuniary savings are meaningful in household budgets. Households with lower income and wealth levels have greater marginal utility of consumption and devote a larger share of their expenditures to energy. As a result, an identical dollar reduction in electricity expenditure has larger welfare consequences for poorer households. To discipline these heterogeneous effects in the model, I use the 2015 Consumer Expenditure Survey (CES) to calculate the energy expenditure shares of U.S. households by income quintile. The results in Table 2 show that the bottom income quintile devotes 8.5% of total expenditure to energy, compared with only 3.3% for the top quintile. These differences in energy budget shares will be reflected in the calibration of household heterogeneity in the model.

2.1.2 Social Benefits

Next, I describe the social benefits of adopting solar panels for residential on-site power generation. I will focus on the environmental benefits of reducing emissions of greenhouse gases and local air pollutants. For example in the US, the residential sector accounted for 15% of end-use energy consumption 2023 according to US Energy Information Administration

(2024) and 19.1% of CO₂ emissions from fossil fuel combustion in 2022 including indirect emissions from electricity according to US Environmental Protection Agency (2024).

The residential sector is also a significant source of air pollutants regulated by the Environmental Protection Agency (EPA) using human health-based and environmentally based criteria under the Clean Air Act. Particle pollution, also known as particulate matter (PM), is a mixture of solid particles, such as dust, dirt, and soot, and liquid droplets found in the air. Breathing in particle pollution can be harmful to human health, as it can cause heart attacks, trouble breathing, and cause lung cancer and problems with babies.

Smaller particles, called PM_{2.5}, pose the greatest health risks because they can get deep into the lungs and the bloodstream. In 2020, fuel combustion by the residential sector accounted for 10.9% and fuel combustion by the electric generation sector accounted for 1.3% of PM_{2.5} emissions in the US according to my calculations using US Environmental Protection Agency’s (2023) 2020 National Emissions Inventory (NEI) data. Dennin et al. (2024) estimate that the marginal damage associated with an additional ton of PM_{2.5} emissions in the US to be between \$73,200 and \$133,000 per ton in 2020 dollars. Thus, given the significant share of PM_{2.5} emissions from the residential sector, reducing emissions from this sector could yield health benefits for local communities.

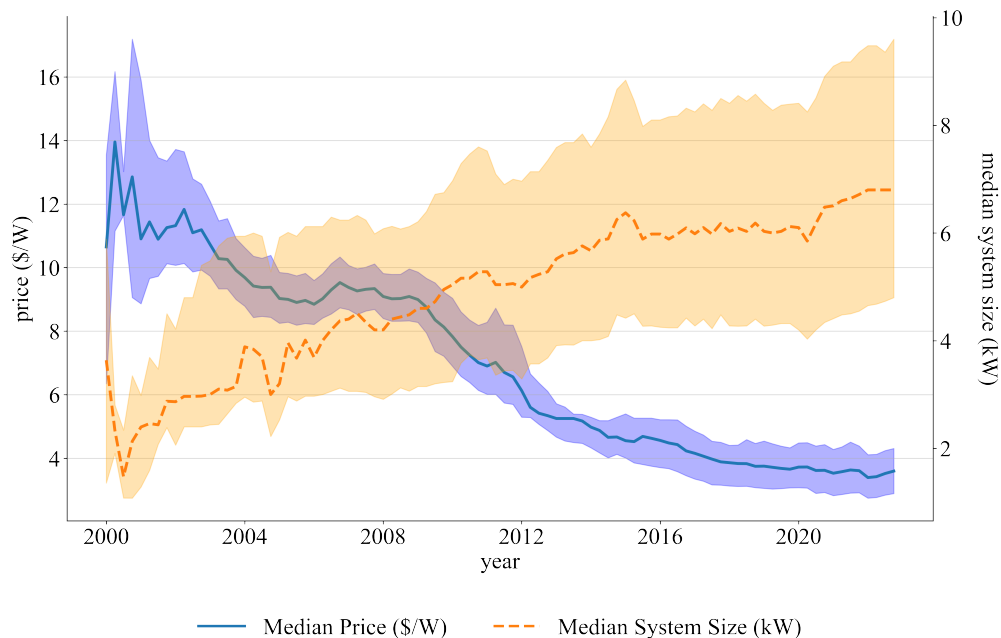
2.2 Cost of Residential Rooftop Solar Panel System Installations

The cost of installing solar panels for on-site power generation is a significant barrier to adoption for many households. The cost of installing solar panels includes the cost of the solar panels, the cost of the inverter, the cost of the mounting hardware, the cost of the wiring, the cost of the installation labor, and the cost of the permitting and inspection. The price of solar panel installations has been decreasing over time due to technological advancements and economies of scale, even before accounting for government incentives.

Figure 1 shows the median price per watt and the median system size of residential solar panels installations in the US from 2000 to 2022. The data is from National Renewable Energy Laboratory’s 2023 2022 Tracking the Sun report data, which covers all non-utility scale solar panel installations in the US. The figure shows that the median price of residential solar panel in the US has declined by almost 65% from 2000 to 2022. The median price per watt declined by almost 65%, while the median system size increased by nearly 75%. This joint trend highlights that, although unit costs fell, households increasingly adopted larger systems, so the decline in total installation costs was slower.

Using these two series on the price and capacity of residential solar installations, I calculate two median total cost of residential solar installation measures in the US from 2000 to

Figure 1: Median price and size of residential solar panel system installations in the US per quarter, 2000-2022



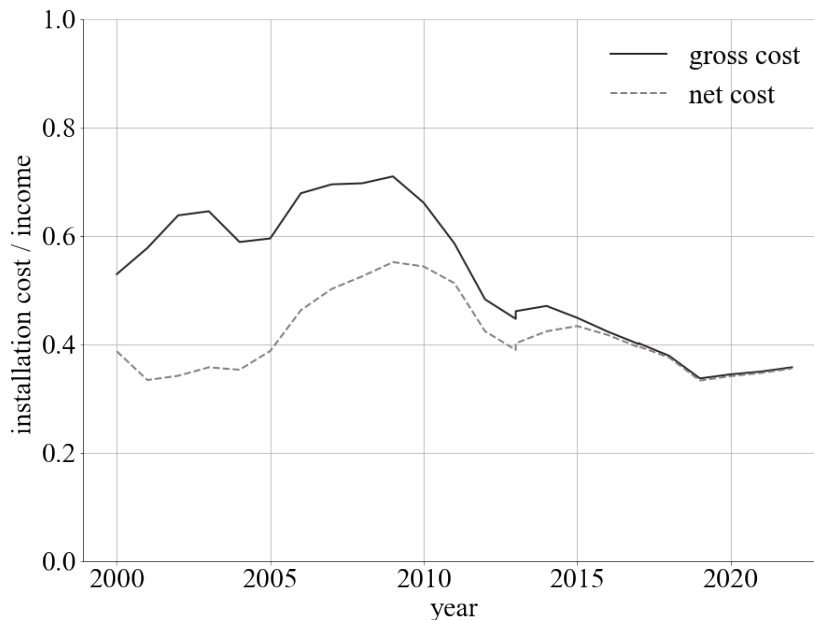
Note: The shaded area represents the 25th and 75th percentiles of the distribution of prices of residential solar panel installations.

2022. The first measure is the median gross total cost, which is the product of the median total installation price per watt and the median system size. The second measure is the median net total cost, which accounts for the state- and utility-level incentives and rebates deducted from the gross total cost for residential solar installations. Importantly, the net total cost measure does not account for the federal investment tax credit (ITC) for residential solar installations, which is 30% of the gross total cost in 2022.

To assess affordability, Figure 2 reports the ratio of these cost measures to median household income in the US, using data from the US Census Bureau’s (2022) ACS 5-Year Estimates (Table S1901). The figure shows that the median gross cost of a residential solar installation fell from about 65% of median household income in 2000 to around 36% in 2022. The gap between gross and net costs narrowed over time and eventually disappeared, reflecting the expiration of many state- and utility-level support programs during this period.

The decline in residential solar installation costs is widely attributed to learning effects. As more systems were produced and installed, both manufacturing and installation processes became more efficient, resulting in lower prices over time.

Figure 2: Ratio of median gross and median net system prices of residential solar PV systems to median annual household income



2.3 Did Rooftop Solar Panel Installations Experience Learning Effects?

The concept of learning effects suggests that as the cumulative experience with a technology increases, the costs associated with that technology tend to decrease. The potential learning effects from the increased adoption of technologies with positive externalities, such as rooftop solar panel installations, can lead to lower costs for future adopters. Thus, subsidies to early adopters of such technologies can have positive spillover effects on the costs of future adopters and has been used to justify public funding for subsidizing early adopters. In the context of rooftop solar panel installations, this could manifest in several ways, including improved manufacturing processes, better installation techniques, and more efficient supply chains.

There are several empirical studies that have found evidence of learning effects in rooftop solar panel installations. Nemet, O'Shaughnessy, et al. (2016) find that more experienced installers consistently quote lower prices than novice installers, all else equal. For instance, low-priced PV systems (the cheapest 10% of installs) are disproportionately completed by installers with extensive prior installations, highlighting how accrued experience translates into cost efficiency. O'Shaughnessy (2018) observed that in more concentrated markets, average installation costs tend to be lower, presumably because high-volume installers climb

farther down the learning curve and achieve economies of scale. However, if a market becomes too concentrated, competitive pressures may diminish, leading to lower prices in moderately concentrated markets that balance scale benefits with competition. Nemet, Lu, et al. (2020) find knowledge spillovers between firms within a county to be a significant and substantial factor in reducing the costs of solar PV installations using data from 2008 to 2014. They find that these spillovers reduce costs only for firms over a certain size threshold and geographic spillovers within a firm across counties to be also significant but smaller than the local between-firm effects. Overall, the literature strongly links cumulative installation experience with cost reductions: every doubling of US residential PV capacity has historically yielded a predictable price decline (a “learning rate”), though estimates of the learning rate for soft costs vary from study to study. Solar Energy Industries Association (2025) reports that the typical residential solar PV installation prices almost halved since early 2010s. Bollinger and Gillingham (2023) estimate that localized learning-by-doing accounted for a modest but measurable decline in installation prices – on the order of \$0.12 per watt reduction in soft costs from each doubling of installer experience in California’s early rooftop solar market data. They also found only limited spillovers between firms, meaning much of the learning was internal to each installer’s operations. On the other hand, Gao, Rai, and Nemet (2022) find that traditional learning-by-doing significantly reduced soft installations costs, although its effect is partly masked by other learning mechanisms, such as accounting for installers’ learning-by-searching (e.g., innovation and R&D) and learning-by-interacting (e.g., knowledge spillovers via supplier networks). Their findings suggest that the traditional learning-by-doing effect may not be as dominant as previous literature suggests.

I test the hypothesis that the cost of residential solar panel installations has decreased over time due to learning effects by regressing the median price of residential solar panel installations on the cumulative installed capacity of residential solar panels in the US and quantify the learning-by-doing effect. There are a variety of statistical models for learning effects all based on the power law of learning, which states that the cost of a technology decreases by a constant percentage with each doubling of cumulative production. I implement the learning-by-doing effect as an exponential decay function, which is a common approach in the literature. The underlying model of learning I assume is as follows:

$$p_t = p_0 \cdot \mathcal{I}_t^{-\xi} \cdot \exp(-\lambda t), \quad (1)$$

where p_t is the net installation price of residential solar panels per watt installed capacity at date t once rebates and incentives are taken into account, p_0 is the initial installation price of residential solar panels per watt installed capacity, $\mathcal{I}_t$ is the cumulative installed capacity

of residential solar panels before date t , could either be a count or a size measure, ξ is the learning-by-doing parameter that captures the rate of cost reduction with each doubling of cumulative installed capacity, and λ is the rate of exogenous time decay that captures the secular declines in PV costs due to global supply chain improvements, technological change, and other factors unrelated to local learning-by-doing. The learning rate implied by this model is given by $1 - 2^{-\xi}$, which represents the percentage reduction in cost associated with each doubling of cumulative installed capacity.

To allow for learning at multiple margins, I include cumulative experience at both the state and firm levels. Let $s(i)$ denote the locality of installation i and $f(i)$ the responsible installer firm. I estimate this model by regressing the natural logarithm of the installation price of residential solar panels on the natural logarithm of the cumulative installed capacity of residential solar panels. Specifically, I use the following regression model:

To accommodate learning at multiple margins, I specify the following regression model:

$$\log p_i = \alpha_{j(i)} - \xi^{\text{state}} \log(\mathcal{I}_{s(i), t(i)-12}^{\text{state}}) - \xi^{\text{firm}} \log(\mathcal{I}_{f(i), t(i)-12}^{\text{firm}}) - \lambda t(i) + \gamma g_{s(i), t(i)} + X_i' \theta + \varepsilon_i, \quad (2)$$

where, p_i is the net installation price per watt of installation i ; $\alpha_{j(i)}$ are fixed effects for the relevant entity j (state, county, or firm) depending on the specification; $\mathcal{I}_{s(i), t-12}^{\text{state}}$ is the cumulative installed residential capacity in state $s(i)$ lagged 12 months; $\mathcal{I}_{f(i), t-12}^{\text{firm}}$ is cumulative installed capacity by firm $f(i)$ lagged 12 months; ξ^{state} and ξ^{firm} are the corresponding learning elasticities; $t(i)$ is the installation month; λ is the exogenous time-decay parameter; $g_{s(i), t(i)}$ measures contemporaneous incentive generosity in the state of installation; X_i is a vector of installation-level controls (system size, hardware, financing); and ε_i is the error term.

In addition to the OLS estimates of equation 2, I consider Instrumental Variable (IV) approaches to address potential endogeneity concerns. Cumulative installations are likely endogenous due to simultaneity bias (e.g., areas or firms with lower costs may attract more installations, making it look like learning when in fact it is selection bias). Moreover, policy support, local demand shocks, or installer entry and exit could drive both installations and prices. An IV approach could isolate the variation in installations that comes from the exogenous policy changes. Finally, an IV approach could provide external validity by leveraging variation across different policy contexts.

I instrument for local cumulative installed capacity using local-level policy changes as an instrument. To capture the exogenous variation in incentives, I construct a monthly binary shock variable that takes the value of 1 if a new residential solar policy starts in location j in month t , and 0 otherwise. To construct this policy shock variable, I use the

North Carolina Clean Energy Technology Center’s [2025](#) Database of State Incentives for Renewables & Efficiency (DSIRE) data, which provides detailed information on state and local policies and incentives for renewable energy and energy efficiency in the US. I provide a detailed description of the construction of this instrument and the policy generosity measure in Appendix [A.1.1](#).

The IV design follows a standard two-stage least squares (2SLS) approach. The core concern is that cumulative installed capacity at the state or county level may be endogenous: areas with falling prices for other reasons may also attract more installations, biasing OLS estimates downwards. I use lagged policy shocks as an instrument for local cumulative installed capacity, to capture the introduction of new state or county incentives (rebates, grants, tax credits, net metering measures), lagged by 12 months to ensure that they affect the stock of cumulative installations but not contemporaneous prices directly. The first stage regression links cumulative installed capacity (the endogenous regressor) to the lagged policy shock, controlling for firm and time fixed effects. A strong first stage relationship is crucial for the validity of the IV approach, as it ensures that the instrument is correlated with the endogenous regressor and satisfy the relevance requirement, and can effectively isolate exogenous variation in installations.

In the second stage, I estimate equation (2) using the predicted values of cumulative installed capacity from the first stage regression, together with additional controls and fixed effects. This approach isolates variation in learning-by-doing that is plausibly exogenous, coming from policy shocks rather than unobserved demand or cost shocks. The exclusion restriction assumes that lagged policy shocks affect installation prices only through their effect on cumulative installed capacity, not directly. This assumption would be reasonable if policies were targeted at adoption rather than subsidies to system prices conditional on installation. However, the majority of the policy do indeed provide direct financial incentives that affect net installation prices, potentially violating the exclusion restriction.

To ensure validity, I implement a refinement of the IV strategy. Specifically, I exploit policy timing shocks by using lagged indicators for the onset of a new policy as instruments for cumulative installations, while simultaneously controlling for the contemporaneous generosity of all active incentives (e.g., rebate amount per watt). This design ensures that any direct effect of subsidies on prices in period t is absorbed by controls, while variation in the timing of policy introductions (lagged L months) provides exogenous shocks to the cumulative stock. After this refinement, my instruments are indicators for whether a new policy incentive began L months earlier, with contemporaneous generosity always included as a control. Identification therefore relies on the assumption that, conditional on current subsidy generosity and fixed effects, the lagged timing of a policy’s onset influences current instal-

lation prices only by raising the cumulative stock of past installations (learning-by-doing), not by directly lowering system costs at the time of observation.

Thus, the first stage regression is as follows:

$$\log(\mathcal{I}_{s(i),t-12}^{\text{state}}) = \mu_{s(i),t-12} + \pi Z_{s(i),t-12} + \rho g_{s(i),t-12} + \delta t + W_{s(i),t-12}'\theta + u_{s(i),t-12}, \quad (3)$$

where $\mu_{s(i),t-12}$ is a vector of entity fixed effects, $Z_{s(i),t-12}$ is the policy shock variable for entity $s(i)$ lagged by 12 months, $W_{s(i),t-12}$ is a vector of additional controls, and $u_{s(i),t-12}$ is the error term. The second stage regression is similar to equation 2, but using the predicted values of cumulative installed capacity from the first stage regression, and is as follows:

$$\log p_i = \alpha_{j(i)} - \xi^{\text{state}} \log(\widehat{\mathcal{I}}_{s(i),t(i)-12}^{\text{state}}) - \xi^{\text{firm}} \log(\mathcal{I}_{f(i),t(i)-12}^{\text{firm}}) - \lambda t(i) + \gamma g_{s(i),t(i)} + X_i'\theta + \varepsilon_i, \quad (4)$$

where $\widehat{\mathcal{I}}^{\text{state}}$ are the predicted values from the first stage regression. The causal effect of learning is identified by ξ .

Equations (3) and (4) constitute a two-stage least squares (2SLS) design. In the first stage, lagged policy shocks serve as instruments for cumulative installed capacity, exploiting the timing of new incentive introductions as an exogenous source of variation in adoption. I include contemporaneous policy generosity, $g_{j,t}$, in both stages to absorb any direct price effects of currently active subsidies, so that the identifying variation arises only from past policy onsets. I also control for a global time trend and a rich set of technological controls to account for background cost declines and other market dynamics. The second stage then regresses log net prices on the predicted cumulative installations from the first stage, controlling for the same covariates and fixed effects. Under the exclusion restriction, the lagged policy shocks affect current installation prices only through their effect on cumulative adoption, not directly, once contemporaneous generosity and time trends are held constant. The coefficient ξ is thus identified as the causal elasticity of installation prices with respect to cumulative installed capacity which I interpret as the learning-by-doing effect.

Table 3 presents results. Columns (1), (3), (5), and (7) report OLS estimates with different fixed effects specifications, while columns (2), (4), and (6) report IV estimates, alternating OLS–IV within each entity level. State FE appear in columns (1)–(4), county FE in (5)–(6), and firm FE in (7). The alternating structure highlights how coefficient magnitudes shift when moving from OLS to IV specifications. The first-stage F-statistics in the IV columns (2), (4), and (6) exceed 10, indicating strong instruments. The coefficients on lagged cumulative installations at both the state and firm levels are the primary parameters of interest, capturing learning effects.

Table 3: Learning-by-Doing in Residential PV with Exogenous Unexplained Decay: OLS and IV with Fixed Effects

Independent variables	(1) OLS	(2) IV	(3) OLS	(4) IV	(5) OLS	(6) IV	(7) OLS
log Cumulative Installs (Firm, 12m lag)	-0.0073*** (0.0003)	-0.0073*** (0.0003)	-0.0097*** (0.0003)	-0.0097*** (0.0003)	-0.0073*** (0.0003)	-0.0073*** (0.0003)	-0.0110*** (0.0006)
log Cumulative Installs (State, 12m lag)	0.0226*** (0.0022)	0.1027*** (0.0054)	0.0244*** (0.0022)	0.0996*** (0.0055)			-0.0033*** (0.0010)
log Cumulative Installs (County, 12m lag)					0.0145*** (0.0014)	0.1249*** (0.0055)	
t	0.0029*** (6.0e-05)	0.0029*** (4.6e-05)	0.0030*** (5.9e-05)	0.0030*** (4.4e-05)	0.0031*** (4.0e-05)	0.0031*** (3.0e-05)	0.0042*** (3.7e-05)
Policy Generosity	-3.4e-07*** (1.7e-08)	-3.4e-07*** (2.0e-08)	-3.4e-07*** (1.7e-08)	-3.4e-07*** (1.9e-08)	-9.1e-07*** (5.3e-08)	-9.1e-07*** (4.7e-08)	-2.1e-07*** (1.8e-08)
Has DC Optimizer	0.0303*** (0.0022)	0.0303*** (0.0022)			0.0347*** (0.0022)	0.0347*** (0.0022)	0.0105*** (0.0022)
Ground Mounted	0.0229*** (0.0053)	0.0229*** (0.0069)			0.0308*** (0.0054)	0.0308*** (0.0069)	0.0673*** (0.0047)
Has Microinverter	0.0098*** (0.0021)	0.0098*** (0.0024)			0.0109*** (0.0022)	0.0109*** (0.0024)	0.0635*** (0.0023)
Inverter Loading Ratio	0.0418*** (0.0038)	0.0418*** (0.0040)			0.0425*** (0.0038)	0.0425*** (0.0040)	-0.0346*** (0.0036)
log Size	-0.1040*** (0.0016)	-0.1040*** (0.0019)			-0.0897*** (0.0016)	-0.0897*** (0.0020)	-0.0952*** (0.0015)
Has Tracking Bin	-0.3999*** (0.0089)	-0.3999*** (0.0298)			-0.4023*** (0.0089)	-0.4023*** (0.0298)	-0.0833*** (0.0090)
First stage							
Policy Shock (State, 12m lag)		-0.7520*** (0.0070)		-0.7518*** (0.0069)			
Policy Shock (County, 12m lag)						-1.1759*** (0.0141)	
F - Statistic		11,705.12		11,701.16		6,995.60	
Number of observations	874,991	874,991	874,991	874,991	874,991	874,991	874,991
R^2	0.060	0.013	0.053	0.013	0.058	0.008	0.054
Location FE	state	state	state	state	county	county	firm

Notes: Robust standard errors in parentheses. Coefficients with robust standard errors in parentheses; ***, **, * denote $p < 0.01, 0.05, 0.10$. Columns are numbered with OLS and IV alternating. First-stage coefficients appear only under IV columns. “log Cumulative Installs (Firm/State/County, 12m lag)” correspond to $\log(\mathcal{I}_{f(i),t-12}^{\text{firm}})$, $\log(\mathcal{I}_{s(i),t-12}^{\text{state}})$, and $\log(\mathcal{I}_{c(i),t-12}^{\text{county}})$. “Policy Shock (State/County, 12m lag)” equals 1 if a new residential PV incentive began in that entity 12 months earlier.

The results in Table 3 show mixed but broadly supportive evidence for learning-by-doing in residential rooftop solar panel installations. All specifications yield negative and statistically significant coefficients on lagged cumulative firm capacity, suggesting that more experienced firms charge higher prices, consistent with market power rather than learning. Whereas, the state- and county-level cumulative installation coefficients are consistently positive and significant when location fixed effects are included in columns (1)-(6), suggesting that local market experience reduces prices. However, the magnitude and significance of these coefficients vary across specifications, ranging from 0.0145 to 0.1249, corresponding

to learning rate estimates ranging from 1 to 8% cost reduction per doubling of cumulative capacity.

The difference in relative magnitudes of state- and county-level learning effects across the OLS and IV specifications is notable. The state-level learning elasticity is four times larger in the IV specifications in columns (2) and (4) than in the OLS specifications in columns (1) and (3). This suggests that OLS may understate state-level learning effects, potentially due to simultaneity and measurement error biases. Instrumenting for cumulative installations with lagged policy shocks appears to amplify the estimated learning effect, indicating that exogenous policy-driven adoption has a stronger impact on cost reductions than organic market growth. The difference between the OLS and IV estimates is even more pronounced at the county level, where the IV estimate in column (6) is nearly six times larger than the OLS estimate in column (5).

The estimates from the OLS specification with firm fixed effects in column (7) are puzzling: both coefficients on cumulative firm capacity and cumulative state capacity are negative and significant, suggesting that more firm or local experience leads to higher prices. This finding contradicts the learning-by-doing hypothesis, which would predict that increased experience leads to lower prices. This counterintuitive result may reflect unobserved heterogeneity among firms, such as market power or strategic pricing behavior, rather than true learning effects. Firms with more experience may have established reputations or customer bases that allow them to charge higher prices, regardless of their actual cost structures and states with more installations may have higher demand and thus higher prices, confounding the learning effect.

My preferred specification is the IV model with state fixed effects in column (2). This specification balances the need to control for unobserved heterogeneity at the state level while still allowing for meaningful variation in cumulative installations. The IV approach helps address endogeneity concerns, and state-level fixed effects capture important policy and market differences across states. The estimated state-level learning elasticity of 0.1027 implies a learning rate of approximately 7% cost reduction per doubling of cumulative capacity.

The inclusion of an exogenous time decay term (λ) reveals a strong and highly significant negative trend across all models, with magnitudes of -0.0015 to -0.003 per month. This implies secular annual declines in PV costs of roughly 2-4% annually, independent of local learning effects. The policy generosity coefficient (γ) is consistently negative, significant and small, indicating that more generous current incentives are associated with lower prices, as expected. Additional controls behave as expected: larger systems are cheaper per watt, while technology upgrades such as optimizers and microinverters increase costs.

My regression results support the consensus view in the literature that rooftop solar

panel installations have experienced learning effects, and state-level experience and markets matter more than purely local and firm learning. Moreover, exogenous time trends are a major driver of cost declines, reflecting global supply chain improvements and technological change. The contribution of these results to existing literature is to provide estimates for both OLS and IV specifications, confirming that OLS may overstate firm-level learning and that state-level cumulative capacity passes IV robustness.

3 Model

Motivated by these empirical observations, I develop a heterogeneous agent dynamic stochastic general equilibrium (DSGE) model with incomplete markets (i.e., Aiyagari-Bewley-Huggett models, building on Bewley (1977), Huggett (1993), and Aiyagari (1994)), with the inclusion of costly clean energy adoption decisions and environmental externalities. The model has discrete-time and infinite horizon. There is a continuum of households indexed by $i \in [0, 1]$. Consumers supply an exogenous amount of labor, receive wage income, accumulate assets in physical capital, and rent out capital to firms. The labor supply is stochastic and is governed by an idiosyncratic shock. Consumers can self-insure by accumulating assets subject to a borrowing constraint.

Households have preferences over consumption and ambient air pollution. Consumption requires energy use, and energy use is assumed to be an affine function of final good consumption. This assumption implies that the expenditure share of energy is larger for households with lower consumption levels, which is an empirical fact documented in Table 2. Consumers have two energy technology options: old fossil (dirty) energy and new renewable (clean) energy. Energy produced by the two technologies are perfect substitutes for each other. However, energy consumption through the old technology creates ambient air pollution – a flow variable that results in utility damages to consumers – a negative externality to society. Before the clean energy technology is available for household adoption, all households use the old energy technology and the economy is in a steady state, i.e., the initial steady state.

Once the clean energy technology is available, consumers can switch to a clean energy source by deploying the clean energy technology, for example rooftop solar panels, to generate electricity with a one-time investment cost and once they make the switch they thenceforth permanently use the clean energy source. The unit cost of clean energy is lower than the unit cost of dirty energy, but the initial investment cost is very high. Specifically, the clean energy technology investment is lumpy: it requires a large one-time and irreversible upfront cost, and once the investment is made, it cannot be recovered. As the scale of clean energy

adoption increases, the investment cost decreases due to learning-by-doing spillover effects. After a sufficiently long time, all households will have adopted the clean energy technology and the economy will be in a new steady state, i.e., the terminal steady state.

Firms combine capital and labor to produce the final good (used for consumption and investment). The negative externality generated by dirty energy consumption is not internalized by the firms or consumers, but leads to a fraction of output being lost due to damages. The loss of utility will be greater for poorer households due to decreasing marginal utility of consumption. Both firms and consumers act competitively and take prices as given. The government collects labor income taxes and uses the revenue to subsidize the investment cost of adopting clean energy technology and returns any excess tax revenues to consumers lump-sum.

3.1 Consumers

Each household is infinitely lived and has preferences over consumption and ambient air pollution. At time t , a household's individual state is described by a vector z_t defined as:

$$z_t = (a_t, \ell_t, s_t) \in \mathcal{Z},$$

where $a_t \in \mathcal{A} = [0, \infty)$ is the household's risk-free asset holding at the beginning of period t , $\ell_t \in \mathcal{L}$ is the idiosyncratic labor productivity endowment at time t , and $s_t \in \{0, 1\}$ is the household's utilization status of the clean energy technology at time t , where $s_t = 0$ indicates that the household is using the fuel combusting old energy technology and $s_t = 1$ is the new clean energy technology. Define the measurable space $(\mathcal{Z}, \mathcal{B}(\mathcal{Z}))$, where:

$$B(\mathcal{Z}) = B(\mathcal{A}) \times P(\mathcal{L}) \times P(\{0, 1\}),$$

with $B(\mathcal{A})$ being the Borel σ -algebra on $\mathcal{A}$ and $P(\cdot)$ the power set. The cross-sectional distribution of households over the state space at time t is represented by a probability measure $\Phi_t \in \mathcal{M}$, where $\mathcal{M}$ is the set of all Borel probability measures on $(\mathcal{Z}, B(\mathcal{Z}))$. For any measurable set $B \in B(\mathcal{Z})$, $\Phi_t(B)$ is the fraction of households with states in B at time t . I will denote $\Phi_t(B)$ by Φ_t when there is no ambiguity. Aggregate objects are computed as integrals with respect to the invariant cross-sectional measure over states, following Huggett (1993).

Each household supplies one unit of time endowment inelastically to the labor market with labor productivity ℓ_t that follows a finite-state Markov chain with transition matrix $\pi(\ell'|\ell)$ and a unique invariant distribution $\Pi(\ell)$. Households derive utility from consumption

and ambient air pollution according to:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t U(c_t, X_t) \right],$$

where $\beta \in (0, 1)$ is the discount factor, c_t is consumption at time t , and X_t denotes the aggregate ambient air pollution at time t . $U(\cdot, \cdot)$ is a strictly increasing and concave one-period utility function in its first argument, strictly decreasing and convex in its second argument, and $\mathbb{E}_0$ is the mathematical expectation conditioned on the consumer's time-0 information.

The household budget constraint is:

$$c_t + a_{t+1} + \bar{q}_t e_t(1 - s_t) + \underline{q}_t e_t s_t + p_t(1 - \tau_t)S_t = w_t(1 - \tau^\ell)\ell_t + (1 + r_t)a_t + T_t,$$

subject to the borrowing constraint:

$$a_{t+1} \geq \underline{a},$$

where $e(\cdot)$ is the energy consumption function that maps household's consumption to energy demand, w_t and r_t denote the wage and interest rate, respectively, $\bar{q}_t$ and $\underline{q}_t$ are the exogenous unit energy prices under dirty and clean energy technologies, respectively, with $\underline{q}_t < \bar{q}_t$ for all t , $s_t^i \in \{0, 1\}$ is the household's utilization status of the clean technology, where $s_t^i = 0$ indicates that the household is using the fuel combusting old energy technology and $s_t^i = 1$ is the new clean energy technology, $S_t^i \in \{0, 1\}$ is the irreversible binary technology adoption decision, p_t is the one-time clean energy technology adoption cost, τ_t is the uniform tax credit (subsidy) for the clean energy technology investment cost, τ^ℓ is the exogenous labor income tax rate, and T_t is the lump-sum transfer. The borrowing limit $\underline{a} \leq 0$ is exogenous and the same for all households.

The discrete technology adoption choice is the main extension of the model from the standard Aiyagari model. The adoption decision, S_t , is a binary choice that is made at the beginning of each period. The adoption decision is irreversible and households who adopt the clean technology in period t will use the clean technology in all future periods. This irreversible adoption decision can be expressed as the difference between the household's next period utilization status of the clean technology and the current period's utilization status of the clean technology:

$$S_t = s_{t+1} - s_t, \text{ with } s_{t+1} \geq s_t \text{ for all } t.$$

The assumption of irreversibility is a reasonable first step, but future work could consider

extensions such as depreciation and replacement of equipment.

3.2 Producers

There is a continuum of competitive firms (with measure normalized to one) producing the consumption good using capital and labor. The production function is:

$$Y_t = F(K_t, L_t),$$

where Y_t is the final output, K_t and L_t are capital and labor demands, respectively, and $F(\cdot, \cdot)$ is a constant returns to scale production function with inputs K_t and L_t . The representative firm takes the factor prices r_t and w_t as given and maximizes its profits each period.

3.3 Government

The government collects tax revenue by taxing every household's labor earnings at rate τ_t^ℓ , uses a portion of its tax revenue to finance the subsidies outlayed for technology adoption investment at rate τ_t , and returns the remainder of its revenue as a uniform lump-sum transfer T_t to each household in each period t . Thus, the government budget constraint is:

$$\int_{\mathcal{Z}} \tau_t^\ell \ell_t d\Phi_t = \int_{\mathcal{Z}} (T_t + \tau_t p_t S_t) d\Phi_t, \quad \forall t. \quad (5)$$

3.4 Ambient Air Pollution

The ambient air pollution each period, X_t , is determined by the flow of energy consumption during the period with the following mapping:

$$X_t = \Omega \left(\int_{\mathcal{Z}} e(c_t)(1 - s_t) d\Phi_t \right), \quad (6)$$

where $\Omega(\cdot)$ is an increasing function that maps the total period t energy consumption of old energy technology users to ambient air pollution. In summary, the air pollution is a function of the total energy consumption of the population that has not adopted low-carbon energy technology, and dirty energy consumption only creates a negative externality within the same period of its generation.

3.5 Learning-by-Doing Spillover

The one-time investment cost of the clean energy technology, p_t , decreases as the cumulative adoption of the clean energy technology increases due to learning-by-doing spillover effects. As I introduced in subsection 2.3, the learning-by-doing spillover effects can be modeled using a power-law of learning function. Specifically, I assume that the adoption cost is a function of the cumulative adoption of the clean energy technology in the economy before period t , denoted by Z_t , defined as

$$Z_t = \int_{\mathcal{Z}} s_t d\Phi_t, \quad (7)$$

and an exogenous time decay component, as stated in equation (1). Thus, the adoption cost function is:

$$p_t = p_0 \cdot Z_t^{-\xi} \cdot \exp(-\lambda t), \quad (8)$$

where $p_0 > 0$ is the initial adoption cost when the technology is first introduced, $\xi > 0$ is the learning-by-doing parameter that captures the rate of cost reduction with each doubling of cumulative adoption, and $\lambda > 0$ is the rate of exogenous time decay that captures the cost declines unexplained by local learning-by-doing spillover effects, such as global supply chain improvements and technological change.

The learning-by-doing spillover effects create a positive externality to society because as more households adopt the clean energy technology, the adoption cost decreases for all future adopters. Modeling the learning-by-doing spillover allows me to evaluate the progressivity of subsidies in a richer sense than just pecuniary transfers, as subsidies also accelerate the learning-by-doing process and thus, the cost declines for future adopters.

3.6 Feasibility

In equilibrium, the market clearing conditions for the capital and labor markets are:

$$\begin{aligned} K_t &= \int_{\mathcal{Z}} a_t d\Phi_t, \\ L_t &= \int_{\mathcal{Z}} \ell_t d\Phi_t \end{aligned}$$

where the left-hand side of each equation is the total demand for the factor and the right-hand side is the total supply of the factor. Denote the market-clearing quantities of aggregate

capital and labor by K_t and L_t , respectively. The goods market clearing condition is:

$$\int_{\mathcal{Z}} [c_t + \bar{q}e(c_t)(1 - s_t) + \underline{q}e(c_t)s_t + p_t S_t] d\Phi_t = F(K_t, L_t) + (1 - \delta)K_t - K_{t+1},$$

where δ is the depreciation rate of aggregate capital stock.

3.7 Recursive Formulation

The model can be formulated recursively. A household's consumption, saving, and adoption decision is governed by its three individual state variables, z_t . In addition, there is one aggregate state variable, the distribution of households across the individual state variables, Φ_t . The ambient air pollution index is not a state variable because it is assumed to be a flow variable and a function of only the current period's aggregate emissions from dirty energy consumption, and thus, is a function of Φ_t alone. The definition of the initial and terminal recursive and stationary competitive equilibria are provided in the Appendix B.1.

3.7.1 Transitional Dynamics

The main focus of the analysis is the transition between the initial steady state with no clean energy technology available to the terminal steady state with every household utilizing the clean energy technology. During the transition, households face the adoption choice $S_t \in \{0, 1\}$, which is irreversible, such that:

$$s_{t+1} = s_t + S_t \geq s_t.$$

The Bellman equation of a household who has not adopted the clean technology and is in state $(a_t, \ell_t, 0)$ at the beginning of period t is given by:

$$\begin{aligned} V_t(a_t, \ell_t, 0; \Phi_t) = \\ \max \left\{ \begin{aligned} & \max_{c_t \geq 0} U(c_t, X_t) + \beta \mathbb{E}_t \left\{ V_{t+1}[w_t(1 - \tau^\ell)\ell_t + (1 + r_t)a_t + T_t - c_t - \bar{q}_t e(c_t), \ell_{t+1}, 0; \Phi_{t+1}] | \ell_t \right\}, \\ & \max_{c_t \geq 0} U(c_t, X_t) + \beta \mathbb{E}_t \left\{ V_{t+1}[w_t(1 - \tau^\ell)\ell_t + (1 + r_t)a_t + T_t - c_t - \bar{q}_t e(c_t) - p_t(1 - \tau_t), \ell_{t+1}, 1; \Phi_{t+1}] | \ell_t \right\} \end{aligned} \right\} \\ \text{subject to } \Phi_{t+1} = \Gamma_t(\Phi_t), \end{aligned} \tag{9}$$

where $\Gamma_t : \mathcal{M} \rightarrow \mathcal{M}$ is the aggregate law of motion in period t governing the distribution of households across the state variables' tomorrow as a function of the distribution today, and $\mathbb{E}_t$ is the expectation operator conditioned on the consumer's time t information. A

household in state $s_t = 0$ at the beginning of period t will choose to adopt the clean energy technology, i.e., set $S_t = 1$ and be in state $s_{t+1} = 1$ at the beginning of period $t + 1$, if the value of adopting is greater than the value of not adopting, i.e., if the second term in the maximization operator is greater than the first term. In the period that follows the decision to adopt, the household is in state $(a_t, \ell_t, 1)$ at the beginning of period t and have the Bellman equation given by:

$$V_t(a_t, \ell_t, 1; \Phi_t) = \max_{c_t \geq 0} U(c_t, X_t) + \beta \mathbb{E}_t \left\{ V_{t+1}[w_t(1 - \tau^\ell)\ell_t + (1 + r_t)a_t + T_t - c_t - \underline{q}_t e(c_t), \ell_{t+1}, 1; \Phi_{t+1}] | \ell_t \right\},$$

subject to $\Phi_{t+1} = \Gamma_t(\Phi_t)$.

(10)

Definition 1 *Given an initial distribution $\Phi_0 \in \mathcal{M}$, fiscal policies $\tau^\ell, \{\tau_t\}_{t=0}^\infty$, and energy prices $\{\bar{q}_t, \underline{q}_t\}_{t=0}^\infty$, a competitive equilibrium is a sequence of: household value and policy functions $\{V_t, c_t, a_{t+1}, S_t, s_{t+1}\}_{t=0}^\infty$, aggregate factor stocks, $\{K_t, L_t\}_{t=0}^\infty$, prices $\{w_t, r_t, p_t\}_{t=0}^\infty$, government transfers $\{T_t\}_{t=0}^\infty$, ambient air pollution levels $\{X_t\}_{t=0}^\infty$, adoption stocks $\{Z_t\}_{t=0}^\infty$, and distributions $\{\Phi_t\}_{t=0}^\infty \subseteq \mathcal{M}$, such that for all t :*

1. **Household optimization.** *The household's value function V_t solves the household Bellman equation given $(w_t, r_t, p_t, \bar{q}_t, \underline{q}_t, \tau^\ell, \tau_t, T_t, X_t, \Phi_t)$, with policy functions $(c_t, a_{t+1}, s_{t+1}, S_t)$ satisfying the budget and borrowing constraints, and the adoption irreversibility constraint.*
2. **Factor prices.** *Factor prices are determined by the marginal products of capital and labor:*

$$r_t = F_K(K_t, L_t),$$

$$w_t = F_L(K_t, L_t).$$

where F_j is the first order partial derivative of F with respect to its input j .

3. **Government budget constraint.** *Government budget constraint holds with equality:*

$$\int_{\mathcal{Z}} \tau^\ell \ell_t d\Phi_t = T_t + \int_{\mathcal{Z}} \tau p_t S_t(a_t, \ell_t, s_t) d\Phi_t.$$

4. **Ambient air pollution.** *Given the distribution of households Φ_t and the policy function $c_t(a_t, \ell_t, s_t)$, the ambient air pollution level X_t satisfies:*

$$X_t = \Omega \left(\int_{\mathcal{Z}} e(c_t(a_t, \ell_t, 0)) d\Phi_t \right).$$

5. **Cumulative adoption.** Given the distribution of households Φ_t , the cumulative adoption level Z_t satisfies:

$$Z_t = \int_{\mathcal{Z}} s_t d\Phi_t.$$

6. **Adoption cost.** Given the cumulative adoption level Z_t , adoption cost satisfies:

$$p_t = p_0 \cdot Z_t^{-\xi} \cdot \exp(-\lambda t).$$

7. **Market clearing.** The factor markets clear:

$$\begin{aligned} K_{t+1} &= \int_{\mathcal{Z}} a_{t+1}(a_t, \ell_t, s_t) d\Phi_t, \\ L_t &= \int_{\mathcal{Z}} \ell_t d\Phi_t, \end{aligned}$$

and the goods market clearing condition:

$$\begin{aligned} &\int_{\mathcal{Z}} [c_t(a_t, \ell_t, s_t) + a_{t+1}(a_t, \ell_t, s_t) + \bar{q}e(c_t(a_t, \ell_t, s_t))(1 - s_t) + \underline{q}e(c_t(a_t, \ell_t, s_t))s_t - p_t S_t(a_t, \ell_t, s_t)] d\Phi_t \\ &= F(K_t, L_t) + (1 - \delta)K_t - K_{t+1}. \end{aligned}$$

8. **Aggregate law of motion.** The aggregate law of motion Γ_t is induced by the transition probabilities and optimal policies $a_{t+1}(a, \ell, s)$, $S_t(a, \ell, s)$, and is explicitly stated in [Appendix B.3](#).

The model's key innovations are the irreversible binary adoption decision with subsidies and the learning-by-doing spillover effects that endogenously lower adoption costs as cumulative adoption rises. Together, these features allow me to study the welfare implications of the clean energy transition and the progressivity of subsidies in a richer sense than pecuniary transfers alone: subsidies not only redistribute resources, but also accelerate learning-by-doing and thereby reduce costs for future adopters. Furthermore, my model allows for heterogeneous impacts of pollution on households with different income levels, which is another feature to study the progressivity of subsidies. The next section describes the calibration of the model for the quantitative analysis. Standard parameters are taken from the existing literature, while the parameters associated with the novel adoption and learning-by-doing mechanisms are calibrated using microdata and reduced-form estimates.

4 Quantitative Analysis

Having laid out the structure of the model, I now turn to its quantitative implementation. To evaluate the distributional and welfare effects of clean energy subsidies, the model must be calibrated to match salient features of the US economy and the residential energy sector. The quantitative analysis proceeds in two steps. First, I describe the parameterization of functional forms and the calibration of the model parameters, distinguishing between those taken from the macroeconomics literature, those pinned down by empirical moments from household- and installation-level data, and those estimated in my own empirical analysis (such as the learning-by-doing elasticity). Second, I outline the computational methods used to solve the model, in the initial and terminal steady states, and during the transition between them under alternative policy scenarios. I defer the discussion of the pollution preference block and its calibration to Section 6.2, where I revisit the baseline results with pollution preferences activated.

I make functional assumptions for the household's utility function, the final goods production function, the dirty and clean energy production functions, the pollution function, and the pollution damage function. I assume that the household's preferences are represented by a constant relative risk aversion (CRRA) utility function of the form:

$$u(c, X) = \frac{c^{1-\sigma} - 1}{1 - \sigma} - \nu \frac{\max\{0, X - \bar{X}\}}{(c/\bar{c})^\omega},$$

where $\sigma > 0$ is the coefficient of relative risk aversion for consumption, $\nu > 0$ scales ambient pollution to utility units, $\bar{X}$ is the pollution threshold above which pollution starts to cause utility losses, $\omega > 0$ makes damages to amplify at lower consumption levels, and $\bar{c}$ is a reference consumption level used to normalize the pollution damage term.

Energy consumption function is an affine function of goods consumption with a nonzero intercept term, with parameters η_0 and η_1 :

$$e(c) = \eta_0 + \eta_1 c,$$

where $\eta_0 > 0$ captures the baseline energy consumption that is independent of goods consumption level, such as energy used for grid connection, while η_1 captures the marginal energy consumption associated with additional final goods consumption. The goods production function is of Cobb-Douglas form:

$$F(K, L) = AK^\alpha L^{1-\alpha},$$

where α is the output share of capital, and A is total factor productivity.

The ambient air pollution function is assumed to be linear in aggregate dirty energy consumption:

$$\begin{aligned} X &= \Omega \left(\int_{\mathcal{Z}} e(c_t)(1 - s_t) d\Phi \right) \\ &= \gamma \left(\int_{\mathcal{Z}} e(c_t)(1 - s_t) d\Phi \right), \end{aligned}$$

where γ is the pollution intensity parameter of dirty energy consumption.

4.1 Calibration

The parameters disciplining the model to match the US economy can be divided into three categories: (i) standard macroeconomic parameters; (ii) parameters pinned down to match moments from data; and, (iii) estimated parameters. One period in the model corresponds to one year, and the initial steady state corresponds to the US economy in 2000, before the widespread adoption of residential solar panels. I summarize the baseline calibration of the structural parameters in Table 4 and provide details below.

4.1.1 Calibration for the Baseline Economy

In the baseline quantitative analysis, I shut down the pollution preference block in utility. Household utility is $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ and ambient pollution X does not affect utility in the baseline. This choice isolates the adoption, affordability, and general equilibrium channels of the residential solar transition.

Standard Macroeconomic Parameters

I begin by assigning values to the set of standard macroeconomic parameters that are commonly used in the heterogeneous agent macroeconomics literature. I follow Aiyagari (1994) and set the output share of capital, α , to 0.36, the discount factor, β , to 0.96, and the capital depreciation rate, δ , to 0.08. I set the total factor productivity of goods production, A , to 1. I assume that the idiosyncratic labor endowment process, ℓ_t , follows a persistent autoregressive process with a persistence parameter of ρ and a standard deviation of σ_ε :

$$\log(\ell_t) = \rho \log(\ell_{t-1}) + \sigma_\varepsilon \sqrt{1 - \rho^2} \varepsilon_t,$$

where $\varepsilon_t \sim \mathcal{N}(0, 1)$. I discretize this earnings process using the Tauchen method. I parameterize this AR(1) labor productivity process with persistence parameter $\rho = 0.9$ and

Table 4: Calibration summary and data sources

Parameter	Description	Value	Source
Standard macroeconomic parameters			
α	Output share of capital	0.36	Aiyagari (1994)
β	Discount factor	0.96	Aiyagari (1994)
δ	Capital depreciation rate	0.08	Aiyagari (1994)
σ	CRRA parameter (final goods consumption)	1	Aiyagari (1994)
ρ	Persistence of labor productivity process	0.9	Aiyagari (1994)
σ_ε	Std. dev. of labor productivity shocks	0.04	Aiyagari (1994)
$\underline{a}$	Borrowing limit	-0.5	PSID (2000)
A	Total factor productivity	1	normalization
Moment matching parameters			
$\bar{q}$	Unit energy price without solar panels	0.04	RECS (2023)
$\underline{q}/\bar{q}$	Unit energy price with solar panels	0.5	RECS (2023)
θ	Fraction of households that can adopt	0.001 $\rightarrow$ 0.15 (linear)	RECS (2023)
τ^ℓ	Uniform labor income tax rate	0.1953	IRS (2000)
τ	Subsidy rate for solar adoption cost	0.3	Lane (2025)
$\pi_{p/y}$	Initial investment cost to income ratio	0.529	NREL (2023) & Census (2024)
Estimated parameters			
ξ	Learning-by-doing (LbD) elasticity	0.1027	NREL (2023) & DSIRE (2025)
λ	Exogenous cost decay parameter	0.0213	NREL (2023) & DSIRE (2025)
η_0	Energy consumption function constant	0.87	BLS (2024) & RECS (2023)
η_1	Energy consumption function slope	0.74	BLS (2024) & RECS (2023)
Pollution block (deferred; not used in baseline)			
γ_ℓ	Mapping dirty energy $\rightarrow$ PM _{2.5}	value from baseline identity	Baseline PM _{2.5} match
$\bar{X}$	PM _{2.5} threshold	9 (robustness: 12)	EPA health standard
$\bar{c}$	Reference consumption	median baseline c	Model moment
ν	Scale (MWTP match)	<i>deferred</i>	BenMAP/VSL + C-R
ω	Inequality lever	<i>deferred</i>	Distributional target

innovation standard deviation $\sigma_\varepsilon = 0.04$ as in one of the parameter combinations considered by Aiyagari (1994). Preferences over consumption are represented by a constant relative risk aversion (CRRA) utility function with a coefficient of relative risk aversion, σ , equal to 1, corresponding to log-utility.

Agents face a non-state-contingent borrowing constraint $a' \geq a$. The theoretical benchmark is the natural borrowing limit (NBL), defined as the present value of the lowest realizable future labor income stream under the no default (solvency) condition; see, Aiyagari (1994).¹ Under my baseline parameters this implies $\underline{a}^{\text{NBL}} \approx -5$ in consumption units. Using the NBL as the operative constraint, however, leads to unstable dynamics and excessive borrowing on a finite grid in this environment (large mass at the constraint and slow convergence along transitions). Following standard practice in incomplete-markets models, I

¹In a stationary environment with net interest rate r and a lower bound on labor income $\underline{\ell}$, the natural borrowing limit is given by $\underline{a}^{\text{NBL}} = -\sum_{s=1}^{\infty} \frac{w \underline{\ell}}{(1+r)^s}$, so that $\underline{a}^{\text{NBL}}$ is the present value of the minimum feasible earnings path.

therefore adopt a tighter, empirically plausible limit of $\underline{a} = -0.5$, which corresponds to an average debt-to-income ratio of approximately 30% in the initial steady state, consistent with the average US household debt-to-income ratio of 30% in 2000 PSID. As robustness, Appendix 6.3 reports results for alternative values $\underline{a} \in (-1, 0]$.

Moment Matching Parameters

The second set of parameters is disciplined by empirical moments from household energy consumption, residential solar panel installation data, and macroeconomic aggregates. Energy prices are pinned down to match the average unit electricity prices reported in the RECS (2023). In particular, the unit electricity price without on-site solar power generation is normalized to $\bar{q} = 0.04$ (in 2020 US dollars per BTU), while households without on-site solar generation face an effective price that is 50% lower, $\underline{q} = 0.5\bar{q}$, reflecting the reduction in electricity consumption from the grid due to on-site solar generation as described in Appendix Table 6. I calibrate the fraction of households that can adopt the clean technology, θ , to 0.5% in the first transition period and linearly increase it to 15% by the terminal steady state, in order to match the observed slow diffusion path of residential solar panels in the US between 2000 and 2020 observed in the RECS (2023).

The uniform labor income tax rate τ^ℓ is set to 19.53%, which corresponds to the revenue-neutral flat tax rate that raises the same total tax revenue as the US federal progressive income tax schedule in 2000, when applied to the model’s steady-state income distribution. The progressive schedule is taken from the Internal Revenue Service’s (IRS) published 2000 tax brackets and rates.² Differences between empirical effective tax rates reflect differences in tax base definitions and sample coverage between the model and the data. For reference, the Congressional Budget Office reports an average effective federal labor income tax rate of 12% in 2000; the gap with the model’s 19.53% reflects differences in the tax base.³ For comparison, I also specify a progressive labor income tax schedule, denoted $\tau^\ell(y)$, that reproduces the US federal marginal tax brackets in 2000 described in Appendix Table 8: This progressive tax specification is used in policy experiments to evaluate the implications of progressivity in the labor income tax system for adoption incentives and welfare. The subsidy rate for residential solar panel adoption cost, τ , is set to 30%, reflecting the average federal investment tax credit (ITC) rate for residential solar installations in the US between 2006 and present as summarized in the Lane (2025). Following the 2025 One Big Beautiful Bill, the ITC rate is set to be eliminated at the end of 2025.

To calibrate the initial investment cost of solar panels, p_0 , I use the ratio of median gross

²<https://www.irs.gov/pub/irs-prior/i1040tt--2000.pdf>

³<https://www.cbo.gov/sites/default/files/108th-congress-2003-2004/reports/08-29-2003AverageTaxRates.pdf>

system price of residential solar panel systems to median annual household income in 2000 and set p_0 such that the model reproduces this ratio internally. I obtain the median gross system price of residential solar panel system installations in 2000 from NREL (2023), and the median annual household income in 2000 from the Census (2024). The time series of this ratio between 2000 and 2022 is plotted in Figure 2. I denote this ratio by $\pi_{p/y}$, which is approximately 52.9 % in 2000, and set the initial investment cost of solar panels as:

$$p_0 = \pi_{p/y} \times \bar{y}_0,$$

where $\bar{y}_0$ is the median annual household income in the initial steady state of the model. Calibrating p_0 in this way ensures that the model matches how adoption decisions are often framed in practice, i.e., in terms of the affordability of the upfront investment cost relative to annual income. These parameters ensure that the model reproduces both the affordability and heterogeneity of residential solar panel adoption in the US.

Estimated Parameters

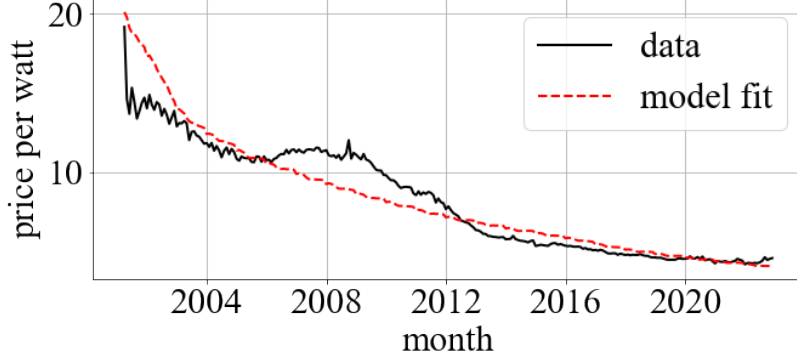
The final set of baseline model parameters is estimated using reduced-form evidence developed in section 2.3 and moments from household energy consumption data. The key parameter of interest is the learning-by-doing elasticity, ξ , which governs how cumulative adoption reduces subsequent installation costs. The elasticity is estimated using state-level panel regressions of residential solar installation prices net of subsidies on local cumulative installed capacity, drawing on data from NREL (2023) and DSIRE (2025) policy database. In addition, I calculate an exogenous time decay parameter, λ , to capture the declines in average costs unrelated to local learning spillovers, such as global technology improvements, economies of scale, and supply chain optimizations. Both ξ and λ are central to quantifying the dynamic effects of subsidies and adoption spillovers in the model.

My baseline learning elasticity ξ estimate comes from the state-level IV regression results reported in column 2 of Table 3, which aligns with the model’s national cost curve while mitigating policy and soft-cost endogeneity concerns. The implied elasticity of 0.1027 implies a learning rate $1 - 2^{-\xi} = 0.0687$, meaning that each doubling of cumulative installed capacity leads to an 6.9% reduction in installation costs. Table 3 also provides results from regression specifications without additional controls and firm-level estimates as robustness checks.

To discipline the residual cost trend, I estimate λ from the full time series rather than only two endpoints. Specifically, I use the transformed learning-curve equation:

$$\log(p_m) + \xi \log(Z_{m-12}) = a - \lambda_m m + \epsilon_m,$$

Figure 3: Fitted cost curve from estimated learning-by-doing parameters



where m is months since the initial sample period and Z_{m-12} is the one-year lagged cumulative installed capacity in month m . The capacity-weighted pre-incentive national monthly price series p_m is constructed using the installation-level data from NREL (2023) as follows:

$$p_m^{\text{gross}} = \frac{\sum_{i \in I_t} ppw_{i,m}^{\text{gross}} \cdot w_{i,m}}{\sum_{i \in I_m} w_{i,m}},$$

where I_m is the set of all residential solar panel installations in month m , $ppw_{i,m}^{\text{gross}}$ is the gross price per watt of installation i in month m before incentives, and $w_{i,m}$ is the size of installation i in watts. The prices are adjusted for inflation using the price deflator.

Together with the estimated elasticity ξ , I run an OLS regression of the left-hand side on a constant and a linear time trend to estimate λ . Specifically, I back out λ_m rearranging the estimated learning-curve equation to obtain:

$$\lambda_m = \frac{\log\left(\frac{p_{m_a}}{p_{m_b}}\right) - \xi \log\left(\frac{Z_{m_b-12}}{Z_{m_a-12}}\right)}{m_b - m_a},$$

where m_a and m_b are the first and last months in the sample period, respectively and p_{m_a} and p_{m_b} are the corresponding capacity-weighted average national gross prices of residential solar panel systems in those months. I convert the estimated monthly λ_m to an annual λ by multiplying by 12 and obtain $\lambda = 0.0213$, indicating an average annual cost increase of 2.5% unexplained by local learning-by-doing spillover effects. Figure 3 plots the fitted cost curve implied by the estimated learning-by-doing elasticity $\xi = 0.1027$ and exogenous time decay parameter $\lambda = 0.0213$ against the actual average gross prices of residential solar panel systems between 2000 and 2022.

The per household adoption cost follows a multiplicative index that combines learning-

by-doing and an exogenous time trend. Let $Z^{\text{pre}} > 0$ denote the pre-adoption experience shifter. I force the cumulative adoption stock $Z_t \geq 0$ to be weakly increasing over time, with $Z_0 = 0$ in the initial steady state. Adoption cost is left-continuous in the experience stock to avoid simultaneity with current installations.

Define the effective experience stock

$$\text{eff}_t \equiv Z^{\text{pre}} + Z_{t-1}, \quad \text{eff}_0 \equiv Z^{\text{pre}} + Z_0,$$

and the (unit-free) cost index

$$\text{idx}_t = \left(\frac{\text{eff}_t}{\text{eff}_0} \right)^{-\xi} \exp(-\lambda \tilde{t}), \quad \tilde{t} \equiv \min\{t, 99\}. \quad (11)$$

The cap $\tilde{t} = \min\{t, 99\}$ halts the pure time trend after 100 periods to prevent implausibly low long-run costs. Numerically, I bound idx_t away from zero by $\text{idx}_t \leftarrow \max\{\text{idx}_t, 10^{-12}\}$ to maintain positivity.

Given a baseline level p_0 , the adoption cost path is

$$p_t = p_0 \times \text{idx}_t, \quad (12)$$

so that p_t falls with accumulated experience (learning) and with the exogenous trend. Let δ^{target} denote the average annual decline in adoption costs over 2000-2020, measured from the data, and let Δ_Z be the average per period increment of the normalized cumulative adoption stock over the same window. For fixed (ξ, λ) , I pin down Z^{pre} so that the model's average log change matches δ^{target} . Under a small-step approximation with mean growth in effective experience, this yields

$$Z^{\text{pre}} = \frac{\Delta_Z}{\exp((\delta^{\text{target}} - \lambda)/\xi) - 1}, \quad (13)$$

and I set $Z^{\text{pre}} = \max\{Z^{\text{pre}}, 10^{-6}\}$ in implementation. In the baseline calibration I use $(\xi, \lambda) = (0.1027, 0.0213)$ and the data moments $(\delta^{\text{target}}, \Delta_Z) = (0.0242, 0.032)$ from 2000-2020 to compute Z^{pre} , which is then held fixed throughout the transition computations.

Finally, I parameterize the affine energy consumption function as follows:

$$e(c) = \eta_0 + \eta_1 c,$$

where $e(c)$ is a household's annual energy expenditure, c is the annual consumption expenditure, and η_0 and η_1 are parameters to be calibrated. I estimate, η_0 and η_1 , to fit the average energy expenditure shares by net worth quintiles from the 2000 Panel Study of Income

Dynamics (PSID) to be replicated in the model's initial steady state.

Let $z \in \mathcal{Z}$ denote the household state (e.g., assets and idiosyncratic labor productivity), and let Φ be the associated invariant probability measure on $(\mathcal{Z}, \mathcal{B}(\mathcal{Z}))$. Let $c(z)$ be goods consumption and let $e(c(z); \eta)$ denote energy services as a function of consumption, parameterized by $\eta = (\eta_0, \eta_1)$. The energy price $\bar{q}$ is taken as exogenous and constant in the baseline.

To map model implications to income quintiles, define an income mapping $\iota : \mathcal{Z} \rightarrow \mathbb{R}_+$ and quantile cutoffs $\{\kappa_j\}_{j=0}^5$ such that

$$\Phi(\{z : \iota(z) \leq \kappa_j\}) = \frac{j}{5}, \quad j = 0, 1, \dots, 5,$$

with $\kappa_0 = -\infty$ and $\kappa_5 = \infty$. The quintile sets are then

$$Q_j := \{z \in \mathcal{Z} : \kappa_{j-1} < \iota(z) \leq \kappa_j\}, \quad j = 1, \dots, 5.$$

For each quintile j , the model-implied energy expenditure share is

$$\epsilon_{Q_j}(\eta) = \frac{\int_{Q_j} \bar{q} e(c(z); \eta) d\Phi(z)}{\int_{Q_j} [\bar{q} e(c(z); \eta) + c(z)] d\Phi(z)}. \quad (14)$$

Let $\hat{\epsilon}_{Q_j}$ denote the empirical targets from the 2000 PSID. The calibration chooses η to minimize the weighted sum of squared deviations:

$$\eta^* \in \arg \min_{\eta \in \mathbb{R}^2} \sum_{j=1}^5 \omega_j \left(\epsilon_{Q_j}(\eta) - \hat{\epsilon}_{Q_j} \right)^2, \quad (15)$$

where $\omega_j = 1$ by default (equal weighting); when available, I set $\omega_j = 1/\hat{\sigma}_{Q_j}^2$ using the sampling variances from PSID (inverse-variance weighting).

On a finite grid $\{z_m\}_{m=1}^M$ with probabilities $\{\Phi_m\}_{m=1}^M$ (so that $\sum_m \Phi_m = 1$), (14) becomes

$$\epsilon_{Q_j}(\eta) = \frac{\sum_{m=1}^M \mathbf{1}\{z_m \in Q_j\} \Phi_m \bar{q} e(c(z_m); \eta)}{\sum_{m=1}^M \mathbf{1}\{z_m \in Q_j\} \Phi_m [\bar{q} e(c(z_m); \eta) + c(z_m)]}.$$

All objects are evaluated at the stationary distribution Φ used for calibration; along transitions one would replace Φ with the relevant Φ_t . The values of η that minimize (15) are $\eta_0 = 0.87$ and $\eta_1 = 0.74$.

4.1.2 Deferred Calibration for Pollution Damages

For transparency, I outline here how the pollution-damage parameters will be identified and report the external data sources. These parameters are *not* used in the baseline calibration or baseline policy experiments; they are activated only in Section 6.2.

Pollution metric and mapping.

I measure X as population-weighted annual $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$), and map dirty energy to ambient pollution by

$$X_t \equiv \gamma \cdot \left[\int_{\mathcal{Z}} e(c_t)(1 - s_t) d\Phi \right].$$

I set γ to match the observed baseline $\text{PM}_{2.5}$ in year 2000 using the baseline model's implied dirty-energy integral with pollution preferences shut down.

Damage function in utility.

When activated, utility is

$$u(c, X) = \frac{c^{1-\sigma} - 1}{1 - \sigma} - \nu \frac{\max\{0, X - \bar{X}\}}{(c/\bar{c})^\omega},$$

where $\bar{X}$ is set to the health-based annual $\text{PM}_{2.5}$ standard (main: 9; robustness: 12), and $\bar{c}$ is the median consumption in the baseline steady state.

Identification strategy (deferred).

1. *Scale* ν : match a marginal willingness-to-pay (MWTP) moment for a 1 $\mu\text{g}/\text{m}^3$ reduction at the baseline using EPA-accepted C-R functions and a VSL (or a direct MWTP estimate).
2. *Inequality lever* ω : target a distributional statistic (e.g., bottom/top ratio of damage-to-consumption shares at baseline) using group-specific consumption and exposures.
3. *Threshold* $\bar{X}$: fixed at the NAAQS level (main) with robustness at the previous standard.
4. *Reference* $\bar{c}$: baseline median consumption.

Status. Parameters $(\nu, \omega, \bar{X}, \bar{c})$ are reported but *inactive* in the baseline; Section 6.2 activates them and revisits policy results with pollution damages and inequality.

Having established the baseline calibration of the model parameters, the next step is to describe how the model is solved and simulated. The computational procedure involves characterizing the household decision problem under the calibrated environment, solving for the stationary equilibrium of the initial and terminal economies, and then tracing out the transition dynamics in response to policy interventions. In what follows, I outline the numerical methods used to solve the model, describe the construction of both the initial and long-run steady states, and detail how transitional paths are computed under baseline subsidy scenario.

4.2 Computation

The model is solved numerically in three stages: (i) the initial steady state without the clean technology, (ii) the terminal steady state with only the clean technology available, and (iii) the transition path connecting the two. Each stage ensures consistency between individual decisions, aggregate quantities, and market clearing conditions.

In the initial steady state, households do not have access to the clean technology, and the economy settles into a stationary equilibrium given exogenous energy prices and fiscal policies. In the terminal steady state, all households are equipped with the clean energy technology, and the economy again reaches a stationary equilibrium under the new energy price regime.

The transition path is computed under perfect foresight. Given an initial guess for the sequences of aggregates, the model is solved by backward induction on household value functions and forward iteration on the distribution of households. Paths of aggregate capital stock, lump-sum transfers, and the cumulative stock of adopters are updated iteratively until factor and goods markets clear at each point along the path. The equilibrium path thus describes the joint evolution of prices, adoption, and welfare as the economy transitions to the new steady state.

Further computational details—including the recursive formulation, iteration schemes, and convergence criteria—are provided in Appendix C.2. I report the key steady-state moments and transition dynamics in Section 5. The computation of the extended model with pollution preferences follows the same steps as above, with the addition of the pollution preference block in utility and the pollution mapping. In Section 6.2, I report the results of the extended model, re-solve the steady states and transition paths, and revisit policy experiments.

The preceding section laid out the calibration strategy and computational methods necessary to map the model to the data. By aligning key parameters with observed macroeconomic

aggregates, energy expenditure statistics, and installation cost dynamics, the calibration ensures that the model reflects both the technological and distributional features central to the adoption decision. The computational approach, in turn, allows for the characterization of households’ dynamic optimization under incomplete markets, as well as the general equilibrium interactions that shape prices and policy incidence. With this foundation in place, the analysis is now equipped to assess the model’s performance relative to the data and to examine the distributional and welfare implications of policy interventions.

5 Quantitative Results

This section presents the quantitative results of the model. I begin by evaluating the fit of the initial stationary equilibrium, focusing on how well the model reproduces salient features of the joint income-wealth distribution observed in the data. This validation step establishes the credibility of the model as a tool for policy analysis. I then turn to the main policy experiment: the introduction of an adoption subsidy for clean energy technologies. The results highlight how the subsidy reshapes adoption incentives across the income and wealth distribution and trace the resulting welfare differences among heterogeneous households. Together, these findings provide a benchmark for understanding both the equity and efficiency consequences of adoption subsidies in the transition to clean energy even before considering the role of pollution externalities.

5.1 Model Fit

Before analyzing the welfare effects of adoption subsidies, I first evaluate how well the initial stationary equilibrium of the model matches key features of the wealth distribution in the 2000 cross-section of the PSID data. Table 5 compares selected variables across the net worth quintiles in the data and the initial stationary equilibrium of the model. The variables include the share of total income, expenditure, and wealth held by each net worth quintile as well as the ratio of energy expenditure to total expenditure.

The model captures the overall distribution of income, expenditure, and wealth reasonably well. The model slightly overpredicts the share of income and expenditure held by the lower net worth quintiles and underpredicts the share held by the highest net worth quintile, but the discrepancies are relatively small. The model also matches the wealth distribution well, although it underpredicts the wealth share of the highest net worth quintile. This discrepancy is a common challenge in heterogeneous-agent models that could be addressed by introducing additional features such as persistent discount factor heterogeneity across

Table 5: Selected variables across the net worth quintiles from the data vs the initial model stationary equilibrium

NW Quintile	% Share of						% Ratio	
	Income		Expenditure		Wealth		Energy-Expend	
	Data	Model	Data	Model	Data	Model	Data	Model
Q1	10.9	13.3	12.6	13.8	-0.9	0.3	5.9	6.2
Q2	12.2	18.2	15.4	18.4	1.2	6.8	6.0	5.6
Q3	17.8	17.4	19.2	17.4	5.2	12.6	5.8	5.1
Q4	23.1	22.6	23.2	22.4	14.8	25.6	5.1	4.8
Q5	35.9	28.5	29.6	28.1	79.6	54.7	4.5	4.4

Notes: Data is from the 2000 PSID.

households. Finally, the model closely replicates the ratio of energy expenditure to total expenditure across the net worth distribution, indicating that it captures the consumption patterns of households well.

5.2 Baseline Results

In order to assess the welfare implications of the energy transition and the role of adoption subsidies during the transition, I compute the welfare changes as consumption-equivalent variations (CEV) for each household in the economy. I provide the details of the welfare calculation in Appendix C.2. The baseline results focus on the case without pollution in the utility function, to isolate the macroeconomic and adoption mechanisms that drive welfare changes. I discuss the role of general equilibrium effects and learning-by-doing in shaping the welfare outcomes. Finally, I explore how the introduction of an adoption subsidy affects welfare across different income and wealth groups.

Aggregate Dynamics and Learning-by-Doing

First, I describe the aggregate dynamics of the economy in response to the introduction of the clean energy technology. In the long-run, the aggregate capital-labor ratio decreases by 12%, which is entirely driven by the reduction in energy use, in consumption units, due to improved energy efficiency. This decline in the capital-labor ratio corresponds to a 4% decline in the real wage and a 24% increase in the real interest rate. Whereas, the 50% decline in energy price leads to 490% increase in good consumption of the poorest households and 160% increase in good consumption of the richest households. In the long-run, although the aggregate capital, and thus output decreases, aggregate consumption increases because of the large decrease in energy price.

However, the path of the aggregate variables is not monotonic over the transition path.

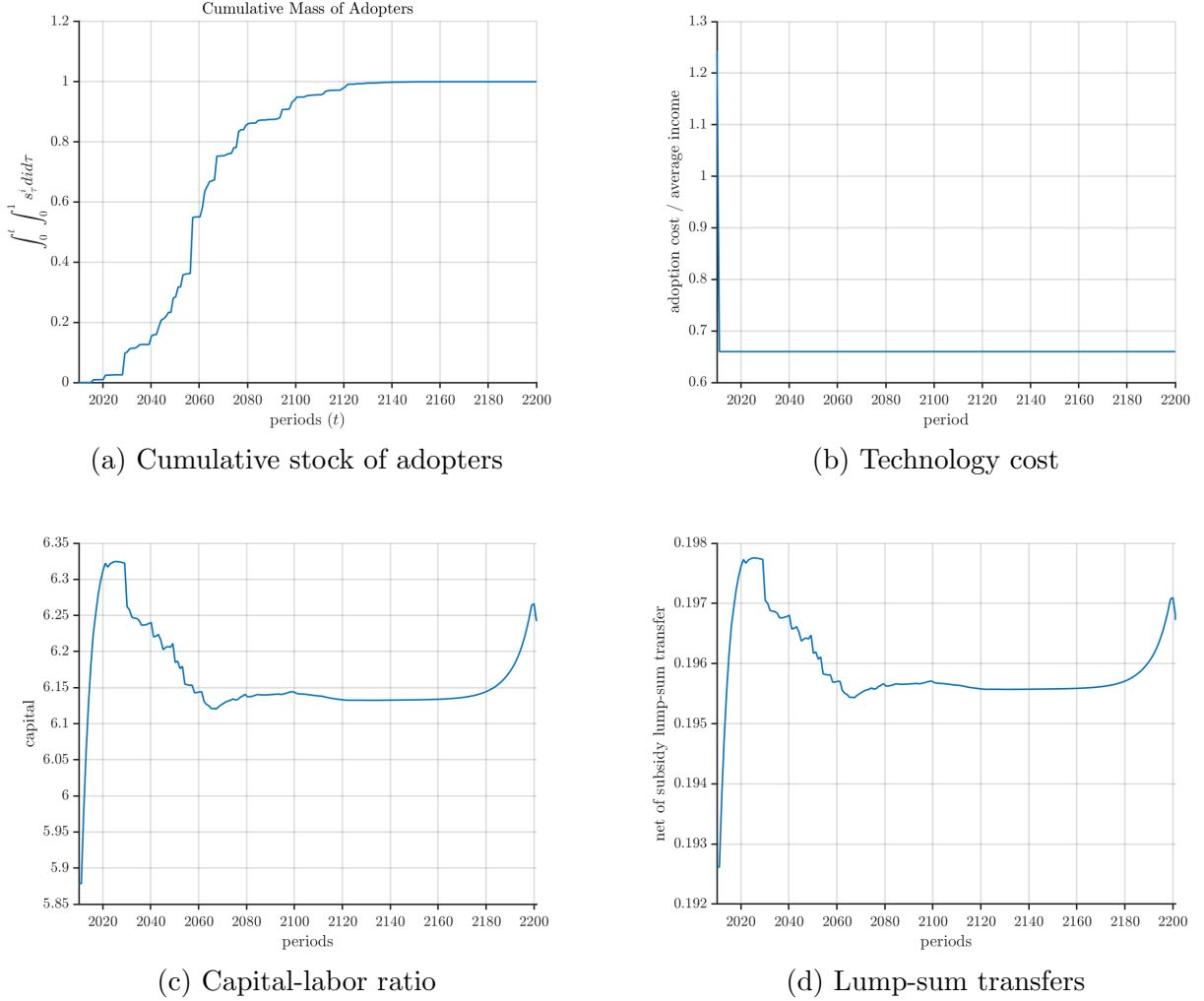
In the short-run, the aggregate capital-labor ratio increases, driven by the saving motive of households to afford the adoption cost of the new technology. This increase in the capital-labor ratio leads to an increase in the real wage over the short-run, which benefits all households, but especially the low-wealth households who rely more on labor income. The real interest rate decreases in the short-run, which hurts savers, especially high-wealth households. As the adoption of the new technology increases over time, the aggregate capital-labor ratio starts to decrease, leading to a decrease in the real wage and an increase in the real interest rate.

The aggregate path of adoption has an S-shape, which is a common feature in the technology adoption literature. The adoption starts slowly, as only high-wealth households can afford the high adoption cost. As the adoption cost decreases due to learning-by-doing (LBD), more households can afford the adoption, leading to a rapid increase in adoption. Finally, as the adoption saturates, the adoption rate slows down. In the short run, as adoption is low and the capital-labor ratio is high, the lump-sum transfers to households increase, which benefits all households, but especially low-wealth households who have higher marginal utility of consumption. As adoption increases (but adoption cost declines) and the capital-labor ratio decreases, the lump-sum transfers decrease, which hurts all households, but especially low-wealth households. Overall, the transition path is characterized by non-monotonic dynamics in the aggregate variables, which have important implications for the welfare of heterogeneous households.

To isolate the role of LBD, I construct a counterfactual in which the installation cost path is fixed to the equilibrium trajectory from the endogenous-learning economy. This ensures that both economies experience the same sequence of cost declines, but only in the endogenous case does the cost evolve as a function of aggregate adoption. Agents remain atomistic and take the cost path as given, so they do not internalize that their own adoption decisions affect future costs. Differences between the two simulations therefore capture the general equilibrium feedback between aggregate adoption and technology costs, rather than differences in the exogenous cost trend itself. The general equilibrium feedback refers to how aggregate adoption affects economy-wide variables (wages, interest rates, fiscal transfers, output), which in turn influence future adoption and welfare.

Figure 4 presents the aggregate dynamics of the economy under the baseline uniform labor income tax policy without adoption technology subsidies, and with and without LBD. The panels show the paths of the cumulative stock of adopters, technology cost, capital-labor ratio, and lump-sum transfers with LBD (solid lines) and without LBD (dashed lines). XXXX

Figure 4: Aggregate dynamics under baseline policy, with uniform labor income taxes without adoption subsidies.

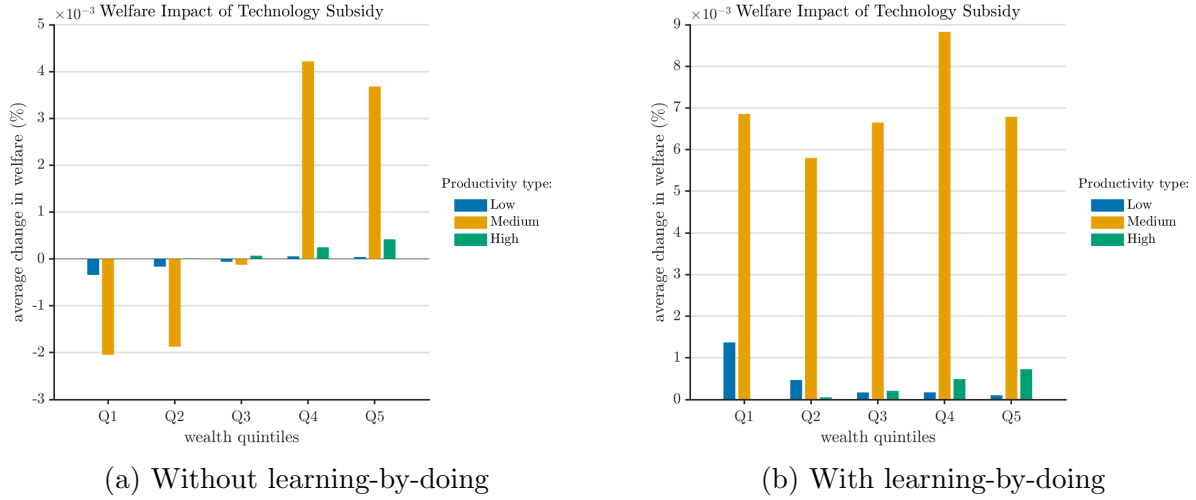


Welfare Distribution Under Baseline Policy

Welfare changes reflect consumption smoothing, price effects, and adoption responses. The pollution preference block is off, so there is no direct disutility from ambient pollution; results isolate pure aggregate adoption mechanisms.

Figure 5 illustrates the distributional consequences of subsidizing the adoption cost of solar panels across wealth quintiles, comparing outcomes with and without LBD, across various labor productivity states at the initial steady state. In the absence of LBD (panel a), welfare gains are concentrated among high-wealth households, particularly those with middle productivity. Low- and middle-wealth households experience negligible benefits, and in some cases modest losses, reflecting their limited ability to adopt even when subsidies are available. This pattern indicates that adoption subsidies operate in a regressive manner

Figure 5: Welfare impact of subsidizing solar panel adoption cost across the income-wealth distribution in general equilibrium



when considered in a static setting.

When LBD is incorporated (panel b), the magnitude of welfare gains increases substantially, but the distributional pattern remains skewed toward wealthier households. High-productivity adopters in the top quintiles capture the largest improvements, while lower-wealth households continue to benefit little, if at all. These findings highlight that dynamic cost reductions amplify the aggregate effectiveness of subsidies but do not resolve their regressive incidence. Instead, they reinforce the concentration of welfare gains among households already well-positioned to adopt, raising equity concerns for subsidy-driven clean energy transitions.

General versus Partial Equilibrium Effects

6 Sensitivity Analysis

Section 6.2 extends the analysis by activating pollution damages in the utility function, allowing me to quantify the additional welfare effects arising from reduced pollution exposure. Finally, Section 6 presents a sensitivity analysis that varies key parameters to assess the robustness of the results.

To move beyond the benchmark subsidy, I evaluate how alternative policy designs affect both the efficiency and equity of the energy transition. Table 8 already shows that simply switching from a uniform to a progressive labor income tax financing mechanism does little to alter the regressive pattern of benefits, particularly in the absence of learning-by-doing.

This motivates the exploration of other financing and subsidy designs.

First, I compare uniform subsidy financing via a flat labor income tax with financing via a progressive labor income tax. Strikingly, progressive financing reduces the overall share of households experiencing welfare gains (Table 8). This occurs because the progressive tax raises costs disproportionately for middle- and low-wealth households who already struggle to adopt, while leaving high-wealth adopters largely unaffected. As a result, subsidies appear more regressive when financed progressively, as fewer low-wealth households benefit, and aggregate welfare gains shrink. This finding highlights an important tension: while progressive taxation is typically viewed as equity-enhancing, in this context it interacts with adoption incentives in a way that exacerbates regressivity.

Second, I shut down adoption subsidies after the first 20 years, to reflect the most recent policy elimination.

Third, I analyze income-capped subsidies that restrict eligibility to middle- and lower-wealth households.

Finally, I compare subsidy financing through debt issuance versus contemporaneous taxation.

Overall, these policy experiments show that no single instrument simultaneously maximizes adoption, equity, and political feasibility. Uniform subsidies financed by debt are efficient but regressive, progressive financing only modestly alters the distribution, and targeted subsidies improve equity at the cost of speed. Policymakers must weigh these trade-offs explicitly.

6.1 Alternative Financing and Subsidy Designs

6.2 Extension: Activating Pollution Damages and Inequality

Calibration Recap

I set γ from the baseline identity, $\bar{X} = 9$ (12 in robustness), $\bar{c}$ to baseline median consumption, identify ν from MWTP at baseline, and set ω to match a distributional burden target. Full details are in Section 4.1.2.

Re-solved Baseline and Policy

With $(\nu, \omega, \bar{X}, \bar{c})$ active, I re-solve the baseline and the policy transitions. I report (i) national welfare including pollution damages, (ii) adoption/welfare by tercile, and (iii) decomposition of gains into market vs pollution channels. A sensitivity analysis varies $\bar{X}$ (9 vs 12), ω ($\pm 25\%$), and VSL bounds.

6.3 Sensitivity Analysis

7 Conclusion

This paper has examined the distributional and welfare consequences of clean technology subsidies in the context of U.S. residential rooftop solar adoption. Using new empirical estimates of learning-by-doing spillovers and a heterogeneous-agent DSGE model with incomplete markets and irreversible adoption, I quantified how subsidies interact with household heterogeneity and dynamic cost declines.

The results show that in a purely static sense, subsidies are regressive: wealthy households adopt first and capture most of the direct benefits. When learning-by-doing is incorporated, aggregate welfare gains increase, but the skewed distribution of benefits persists. Importantly, the incidence of subsidy financing matters: progressive labor taxation—often thought of as equity-improving—actually worsens regressivity in this setting, because it reduces the ability of lower-wealth households to benefit from subsidies. By contrast, uniform tax or debt financing delivers higher aggregate gains, though it raises separate concerns about fairness and intergenerational equity.

The broader implication is that evaluating energy subsidies purely through the lens of static pecuniary incidence understates their impact. Dynamic spillovers mean that subsidies can accelerate cost declines, ultimately enabling broader adoption. Yet absent complementary measures, these benefits remain tilted toward the top of the distribution. Equity in the energy transition therefore requires a portfolio approach: broad early subsidies to exploit learning externalities, combined with targeted or progressive policies to ensure inclusivity.

This framework offers a tractable way to quantify the equity–efficiency trade-offs inherent in clean energy policy. Future work could extend the model to incorporate endogenous energy prices or labor market reallocation. For now, the results suggest that designing subsidies with both efficiency and distributional goals in mind is crucial to building a clean energy transition that is not only fast but also fair.

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Table 6: Regression results for the effect of solar generation on annual electricity expenditure

Variables	Electricitiy		Total Energy	
	(1)	(2)	(3)	(4)
Solar	-0.0214 (0.000)	-0.0214 (0.000)	-0.0093 (0.000)	-0.0091 (0.000)
State Controls	Yes	Yes	Yes	Yes
Additional Controls	Yes	No	Yes	No
Observations	15,044	15,044	15,044	15,044

Notes: Standard errors are in parentheses. Fixed effects indicate the inclusion of state fixed effects and additional controls refer to inclusion of factors such as household and housing unit characteristics.

Table 7: Regression results for the effect of solar generation on annual electricity expenditure

Variables	Electricitiy		Total Energy	
	(1)	(2)	(3)	(4)
Solar	-721.956 (15.938)	-709.430 (15.881)	-688.5617 (24.286)	-609.9241 (27.094)
State Controls	Yes	Yes	Yes	Yes
Additional Controls	Yes	No	Yes	No
Observations	15,044	15,044	15,044	15,044

Notes: Standard errors are in parentheses. Fixed effects indicate the inclusion of state fixed effects and additional controls refer to inclusion of factors such as household and housing unit characteristics.

Appendices

Appendix A Empirical Appendix

A.1 Data Sources and Variable Construction

A.1.1 Policy Shocks Instrument

DSIRE ([2025](#))

Appendix B Theoretical Appendix

B.1 Initial and Terminal Recursive Equilibria

Prior to the availability of the clean energy technology for adoption, households face no adoption decision and are in state $s = 0$. The state space is:

$$\tilde{\mathcal{Z}} = \mathcal{A} \times \mathcal{L}, \quad B(\tilde{\mathcal{Z}}) = B(\mathcal{A}) \times P(\mathcal{L}),$$

and the distribution of households is given by $\Phi \in \tilde{\mathcal{M}}$, where $\tilde{\mathcal{M}}$ is the set of Borel probability measures on $(\tilde{\mathcal{Z}}, B(\tilde{\mathcal{Z}}))$.

Definition 2 *Given the labor income tax rate τ^ℓ , exogenous dirty energy price $\bar{q}$, and ambient air pollution function $X(\Phi)$, a recursive competitive equilibrium consists of: a value function $V : \tilde{\mathcal{Z}} \times \tilde{\mathcal{M}} \rightarrow \mathbb{R}$, household policy functions $c, a' : \tilde{\mathcal{Z}} \times \tilde{\mathcal{M}} \rightarrow \mathbb{R}$, aggregate factor demands $K, L : \tilde{\mathcal{M}} \rightarrow \mathbb{R}$, factor price functions $r, w : \tilde{\mathcal{M}} \rightarrow \mathbb{R}$, a transfer function $T : \tilde{\mathcal{M}} \rightarrow \mathbb{R}$, a pollution function $X : \tilde{\mathcal{M}} \rightarrow \mathbb{R}$, and law of motion $H : \tilde{\mathcal{M}} \rightarrow \tilde{\mathcal{M}}$ such that:*

1. **Household optimization.** *Given the pricing functions $r(\Phi), w(\Phi)$, transfer function $T(\Phi)$, V solves the following Bellman equation:*

$$V(a, \ell; \Phi) = \max_{c \geq 0} U(c, X(\Phi)) + \beta \mathbb{E} \{ V[w(\Phi)(1 - \tau^\ell)\ell + (1 + r(\Phi))a + T(\Phi) - c - \bar{q}e(c), \ell'; \Phi'] | \ell \},$$

subject to $\Phi' = H(\Phi)$,

(16)

and c is the associated consumption policy function, $a' = w(\Phi)(1 - \tau^\ell)\ell + (1 + r(\Phi))a + T(\Phi) - c - \bar{q}e(c)$ is the savings policy function, and $\mathbb{E}$ is the conditional expectation operator.

2. **Factor prices.** *Factor prices $r(\Phi)$ and $w(\Phi)$ satisfy the firm's first-order conditions:*

$$r(\Phi) = F_K(K(\Phi), L(\Phi)),$$

$$w(\Phi) = F_L(K(\Phi), L(\Phi)).$$

3. **Government budget.** *Given the factor prices $r(\Phi)$ and $w(\Phi)$ and the tax rate τ^ℓ , the government runs a balanced budget every period such that $T(\Phi)$ satisfies:*

$$T(\Phi) = w(\Phi)\tau^\ell \int_{\tilde{\mathcal{Z}}} \ell d\Phi.$$

4. **Pollution.** The ambient air pollution $X(\Phi)$ is a function of the aggregate energy consumption and satisfies:

$$X(\Phi) = \Omega \left(\int_{\tilde{\mathcal{Z}}} e(c(a, \ell; \Phi)) d\Phi \right).$$

5. **Market clearing.** For all $\Phi \in \tilde{\mathcal{M}}$,

$$\begin{aligned} L(\Phi) &= \int_{\tilde{\mathcal{Z}}} \ell d\Phi, \\ K(H(\Phi)) &= \int_{\tilde{\mathcal{Z}}} a'(a, \ell; \Phi) d\Phi, \\ \int_{\tilde{\mathcal{Z}}} [c(a, \ell; \Phi) + a'(a, \ell; \Phi) + \bar{q}e(c(a, \ell; \Phi))] d\Phi &= F(K(\Phi), L(\Phi)) + (1 - \delta)K(\Phi) - K(H(\Phi)). \end{aligned}$$

6. Aggregate law of motion H is generated by π and a' , explicitly stated in Appendix B.2.

Next, I characterize the recursive competitive equilibrium in the initial steady state as follows:

Definition 3 Given the labor income tax rate τ^ℓ the stationary recursive equilibrium is a value function V , household policy functions c, a' , aggregate production factors K, L , prices $r, w, \bar{q}$, government transfer T , pollution function X , and a measure Φ , with $\Phi \in \tilde{\mathcal{M}}$ invariant under H , such that the household optimization, factor prices, government budget, pollution, and market clearing conditions above hold, and Φ satisfies:

$$\Phi = H(\Phi).$$

Once adoption is available and irreversible, the terminal steady state has all households in state $s = 1$. Then $\mathcal{Z} = \mathcal{A} \times \mathcal{L} \times \{1\}$. The recursive equilibrium is defined analogously, with $\bar{q}$ replaced by $\underline{q}$ and $X(\Phi) = 0$ for all Φ .

B.2 Explicit Statement of the Aggregate Law of Motion in the Steady States

First, define the Markov transition function: $Q_\Phi : \tilde{\mathcal{Z}} \times B(\tilde{\mathcal{Z}}) \rightarrow [0, 1]$ by:

$$Q_\Phi((a, \ell), (\mathcal{A}, \mathcal{L})) = \sum_{\ell' \in \mathcal{L}} \begin{cases} \Pi(\ell' | \ell) & \text{if } a'(a, \ell; \Phi) \in \mathcal{A}, \\ 0 & \text{otherwise,} \end{cases}$$

for all $(a, \ell) \in \tilde{\mathcal{Z}}$ and $(\mathcal{A}, \mathcal{L}) \in B(\tilde{\mathcal{Z}})$. Thus $Q_\Phi((a, \ell), (\mathcal{A}, \mathcal{L}))$ is the probability that an agent with current assets a and productivity ℓ ends up with assets $a' \in \mathcal{A}$ and productivity $\ell' \in \mathcal{L}$ tomorrow. Then, the aggregate law of motion for the initial steady state distribution is given by:

$$\begin{aligned}\Phi'(\mathcal{A}, \mathcal{L}) &= H(\Phi)(\mathcal{A}, \mathcal{L}) = \int_{\tilde{\mathcal{Z}}} Q_\Phi((a, \ell), (\mathcal{A}, \mathcal{L})) \\ &= \int Q_\Phi((a, \ell), (\mathcal{A}, \mathcal{L})) \Phi(da \times d\ell).\end{aligned}$$

B.3 Explicit Statement of the Aggregate Law of Motion During the Transition Path

Define the Markov transition functions $Q_t : \mathcal{Z} \rightarrow [0, 1]$ induced by the transition probabilities π and optimal policies $a_{t+1}(a, \ell, s)$ and $S_t(a, \ell, s)$ as:

$$Q_t((a, \ell, s), (\mathcal{A}, \mathcal{L}, \{0, 1\})) = \sum_{\ell' \in \mathcal{L}} \begin{cases} \pi(\ell' | \ell) & \text{if } a_{t+1}(a, \ell, s) \in \mathcal{A}, \\ 0 & \text{otherwise,} \end{cases}$$

for all $(a, \ell, s) \in \mathcal{Z}$ and $(\mathcal{A}, \mathcal{L}, \{0, 1\}) \in B(\mathcal{Z})$. Then, for all $(\mathcal{A}, \mathcal{L}, \{0, 1\}) \in B(\mathcal{Z})$, the aggregate law of motion for the transition distribution is given by:

$$\Phi_{t+1}(\mathcal{A}, \mathcal{L}, \{0, 1\}) = [\Gamma_t(\Phi_t)](\mathcal{A}, \mathcal{L}, \{0, 1\}) = \int Q_t((a, \ell, s), (\mathcal{A}, \mathcal{L}, \{0, 1\})) d\Phi_t.$$

Appendix C Quantitative Appendix

C.1 Calibration Details

Table 8: Income Tax Brackets and Rates

Income Bracket (\$2000)	Marginal Tax Rate (%)
0 - 26,250	15%
26,250 - 63,550	28%
63,550 - 132,600	31%
132,600 - 288,350	36%
288,350 - above	39.6%

C.2 Computation Details

Steady States

Two steady states are computed:

1. **Initial steady state:** Clean energy technology is unavailable ($s = 0$), exogenous dirty energy price is $\bar{q}$, and the entire proceeds from the exogenous and flat labor income tax rate τ^ℓ is distributed lump-sum to households equally. After initializing parameters, guess K and T , and solve the Bellman equation (16) using Value Function Iteration with linear interpolation and a uniform asset grid. I use Golden Section Search for optimization and Howard's step for speed improvements. I compute the invariant distribution of (a, ℓ) by iterating over the density function on a finer uniform asset grid until convergence. Finally, I compute the K and T implied by the invariant distribution and compare to the initial guesses, updating using a dampening parameter $a = 0.95$ until convergence within a tolerance threshold equal to 10^{-6} .
2. **Terminal steady state:** The entire population has adopted the clean technology ($s = 1$), the exogenous clean energy price is $\underline{q}$, and the entire proceeds from the exogenous and flat labor income tax rate τ^ℓ is distributed lump-sum to households equally. The same algorithm is used to find $(K_\infty, T_\infty, \Phi_\infty)$.

Transition Path Algorithm

Given a finite horizon $T = 190$, the model is solved under perfect foresight using an outer-loop fixed-point algorithm:

1. *Initialization:* Import the value and policy functions for both the initial and terminal steady state calculations. Also import the invariant distribution at the initial steady state.
2. *Guessing:* Guess sequences $\{K_t, T_t, Z_t\}_{t=1}^T$, where $t = 1$ is the period when the transition starts and $t = T$ is sufficiently far in the future so that I can assume that the economy is sufficiently close to the new steady state. Also calculate the factor prices $\{r_t, w_t\}_{t=1}^T$ implied by evaluating the firm's first-order conditions at the guessed capital path.
3. *Backward induction:* at $t = T + 1$, the economy is in the terminal steady state, so $K_{T+1} = K_\infty$, $T_{T+1} = T_\infty$, and $V_{T+1} = V_\infty$ and I can use the terminal value function from the previous step in the right-hand side of the period- T Bellman equation, given

by equations (9) and (10). From there, I solve for the household's value and policy functions with backward induction for $t = 1, \dots, T$.

4. *Forward simulation:* Using the policy function from the previous step, I simulate the economy forward, starting from the imported density function at the initial steady state. Using the simulated distribution of households, I compute K_t , T_t , and Z_t implied by the distribution at each $t = 1, \dots, T$.
5. *Convergence:* Compare the implied $\{K_t, T_t, Z_t\}_{t=1}^T$ to the initial guesses and modify the guesses using a dampening parameter in $(0, 1)$ until convergence of the guess and updated paths within a tolerance threshold equal to 10^{-6} .

Welfare Change Calculations

For each household state (a, ℓ, s) , welfare changes are computed as consumption-equivalent variations (CEVs) between the baseline and counterfactual paths. The CEV is defined as the percentage change in consumption that, if applied to the baseline consumption path, would yield the same lifetime utility as under the counterfactual path. For example, the welfare change from subsidizing the clean energy technology adoption at 30% is computed as $\lambda(a, \ell, s)$ that solves:

$$\mathbb{E}_1 \left[\sum_{t=1}^T \beta^t U((1 + \lambda(a, \ell, s))c_t(a, \ell, s)^{\tau=0}) \right] = \mathbb{E}_1 \left[\sum_{t=1}^T \beta^t U(c_t(a, \ell, s)^{\tau=0.3}) \right],$$

where $c_t(a, \ell, s)^{\tau=0}$ is the consumption policy function at time t under the baseline scenario with no subsidy and $c_t(a, \ell, s)^{\tau=0.3}$ is the consumption policy function at time t under the counterfactual scenario with a 30% subsidy for a household with state (a, ℓ, s) at $t = 1$. If $\lambda(a, \ell, s) > 0$, then the household is better off under the counterfactual scenario and supports subsidizing the clean energy technology adoption at 30%. Under the CRRA utility specification, the CEV can be computed in closed form as:

$$\lambda(a, \ell, s) = \begin{cases} \exp \{ (1 - \beta) [V_1(a, \ell, s)^{\tau=0.3} - V_1(a, \ell, s)^{\tau=0}] \} - 1, & \text{if } \sigma = 1, \\ \left[\frac{V_1(a, \ell, s)^{\tau=0.3}}{V_1(a, \ell, s)^{\tau=0}} \right]^{\frac{1}{1-\sigma}} - 1, & \text{if } \sigma \neq 1, \end{cases}$$

where $V_1(a, \ell, s)^{\tau=0}$ is the value function at $t = 1$ under the scenario with no adoption subsidy and $V_1(a, \ell, s)^{\tau=0.3}$ is the value function at $t = 1$ under the scenario with a 30% adoption subsidy for a household with state (a, ℓ, s) at $t = 1$.

In order to compute the welfare change due to the introduction of the clean energy

technology, I compute the CEV as:

$$\mathbb{E}_1 \left[\sum_{t=1}^T \beta^t U((1 + \lambda(a, \ell, s))c_t(a, \ell, s)^{\text{initial ss}}) \right] = \mathbb{E}_1 \left[\sum_{t=1}^T \beta^t U(c_t(a, \ell, s)^{\text{transition}}) \right],$$

where $c_t(a, \ell, s)^{\text{initial ss}}$ is the consumption path under no technology transition (the technology remains available and the economy remains in the initial steady state) and $c_t(a, \ell, s)^{\text{transition}}$ is the consumption path under the technology transition, without any subsidies. If $\lambda(a, \ell, s) > 0$, then the household is better off with the introduction of the clean energy technology and supports the transition.